MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL
PHYSICS

Light

3.2

STUDENT NAME

PHYSICS WITH
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O'Levels 3.2

Light

MOIZ DAWOODI
O/A LEVEL PHYSICS

SYLLABUS OUTLINE:

3.2 Light

3.2.1 Reflection of light

1 Define and use the terms normal, angle of incidence and angle of reflection

2 Describe an experiment to illustrate the law of reflection

3 Describe an experiment to find the position and characteristics of an optical image formed by a plane mirror (same size, same distance from mirror as object and virtual)

4 State that for reflection, the angle of incidence is equal to the angle of reflection and use this in constructions, measurements and calculations

3.2.2 Refraction of light

1 Define and use the terms normal, angle of incidence and angle of refraction

2 Define refractive index n as $n = \dfrac{\sin i}{\sin r}$; recall and use this equation

3 Describe an experiment to show refraction of light by transparent blocks of different shapes

4 Define the terms critical angle and total internal reflection; recall and use the equation

$$n = \frac{1}{\sin c}$$

5 Describe experiments to show internal reflection and total internal reflection

6 Describe the use of optical fibres, particularly in telecommunications, stating the advantages of their use in each context

3.2.3 Thin lenses

1 Describe the action of thin converging and thin diverging lenses on a parallel beam of light

2 Define and use the terms focal length, principal axis and principal focus (focal point)

3 Draw ray diagrams to illustrate the formation of real and virtual images of an object by a converging lens and know that a real image is formed by converging rays and a virtual image is formed by diverging rays

4 Define linear magnification as the ratio of image length to object length; recall and use the equation

$$\text{linear magnification} = \frac{\text{image length}}{\text{object length}}$$

5 Describe the use of a single lens as a magnifying glass

6 Draw ray diagrams to show the formation of images in the normal eye, a short-sighted eye and a long-sighted eye

7 Describe the use of converging and diverging lenses to correct long-sightedness and short-sightedness

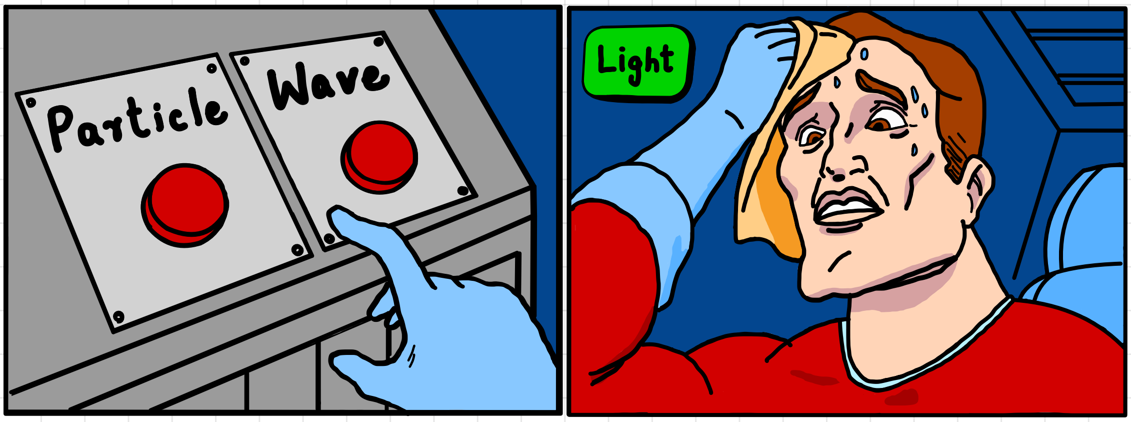
3.2.4 Dispersion of light

1 Describe the dispersion of light as illustrated by the refraction of white light by a glass prism

2 Know the traditional seven colours of the visible spectrum in order of frequency and in order of wavelength

LIGHT:

"Light is a form of energy & a member of electromagnetic spectrum. It can reflect & refract. It travels in a straight line with speed of 3×10^8 m/s in vacuum. It does not require any medium to travel. It helps us to see things around us."



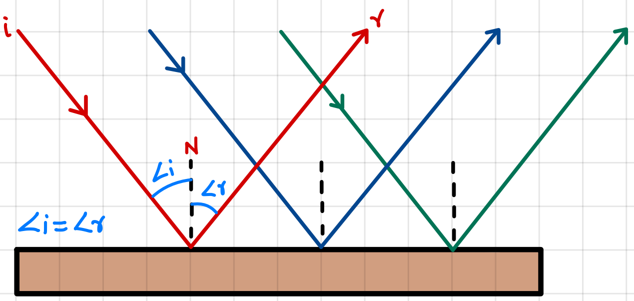
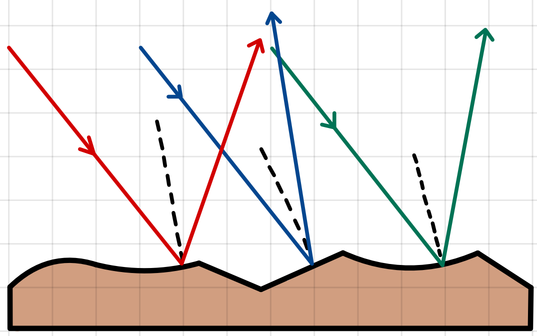
3.2.1 Reflection of Light

REFLECTION OF LIGHT:

"Bouncing back of light into the same medium is called reflection."

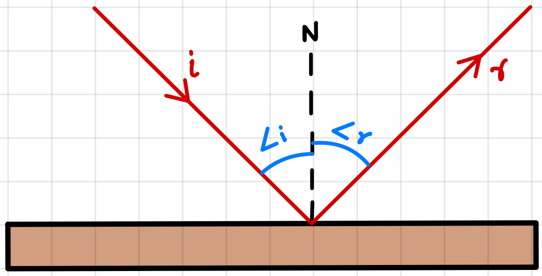
LAWS OF REFLECTION:

- The incident ray, reflected ray & normal lie on same plane.
- Angle of incidence is equal to angle of reflection.

TYPES OF REFLECTION	
REGULAR REFLECTION	IRREGULAR REFLECTION
	

ANGLE OF INCIDENCE:

"The angle b/w incident ray & normal is called angle of incidence.



ANGLE OF REFLECTION:

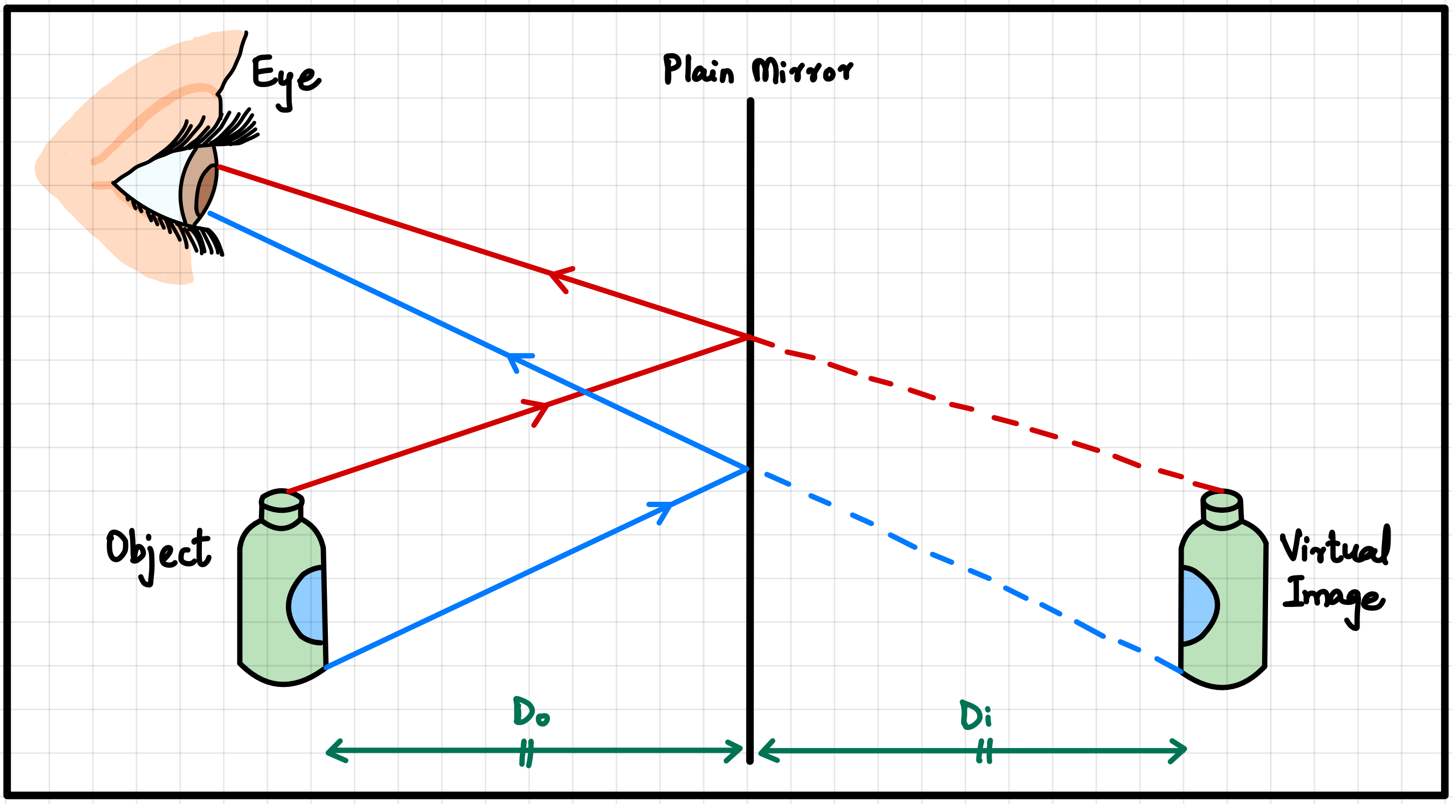
"The angle b/w reflected ray & normal is called angle of reflection.

Imaginary lines are drawn with doted lines while real lines are drawn with solid lines.

NORMAL:

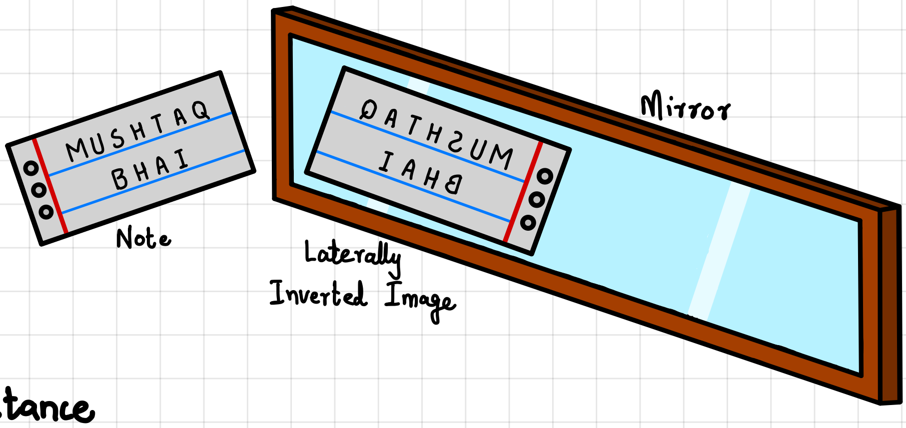
"The imaginary line which makes 90° with the surface is called normal.

Image Formed By Plain Mirror:

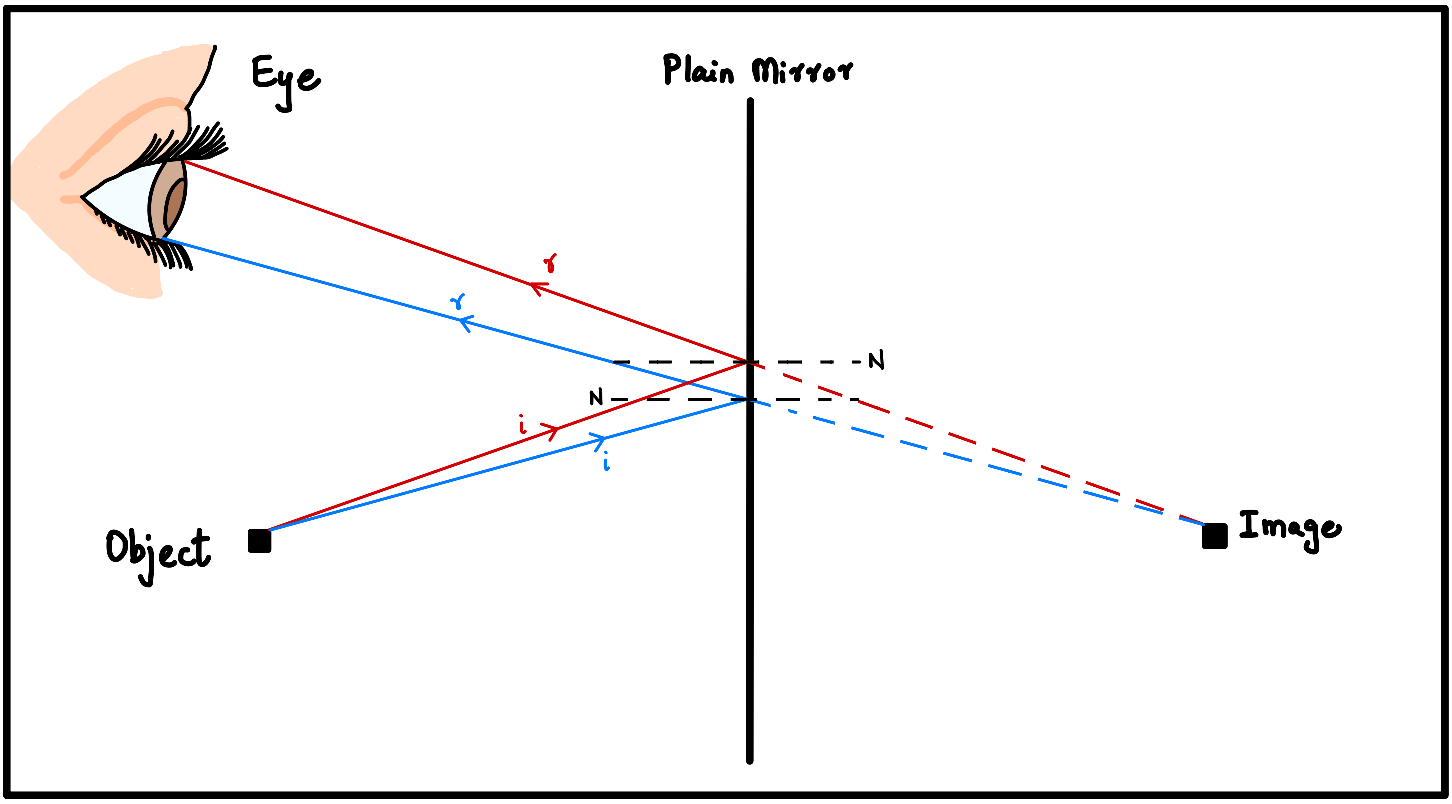


PROPERTIES:

- Virtual Image
- Laterally Inverted
- Object Size = Image Size
- Object distance = Image distance

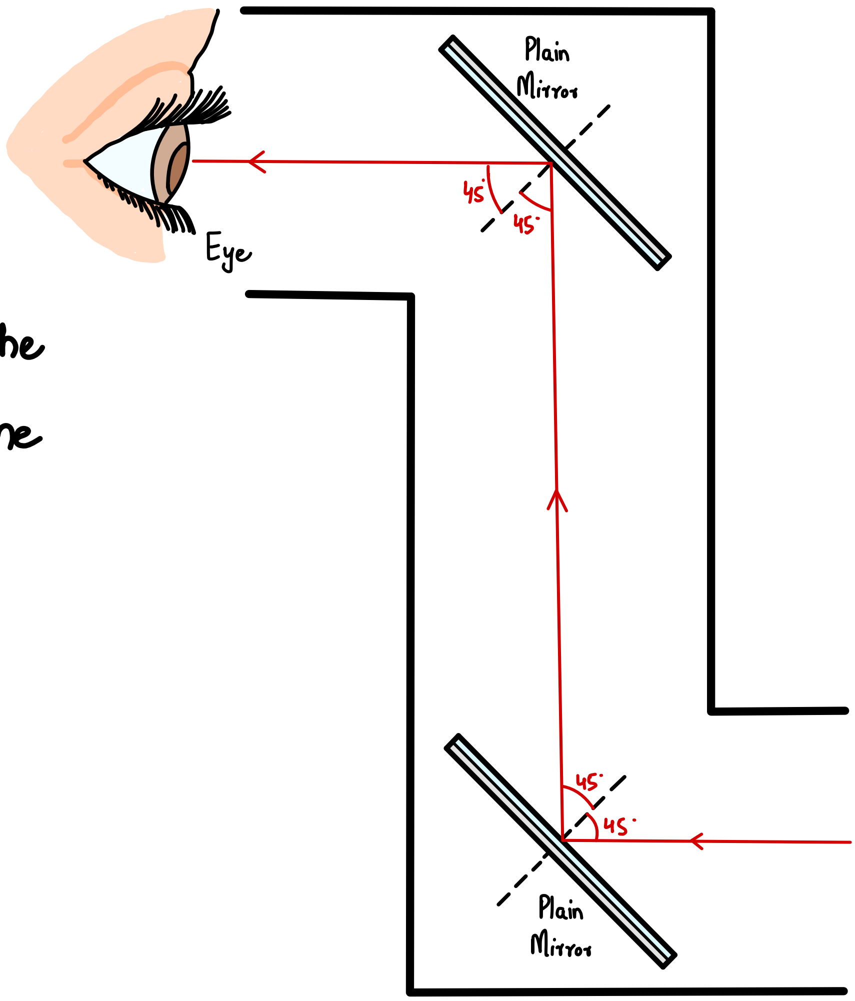


RAY DIAGRAM:

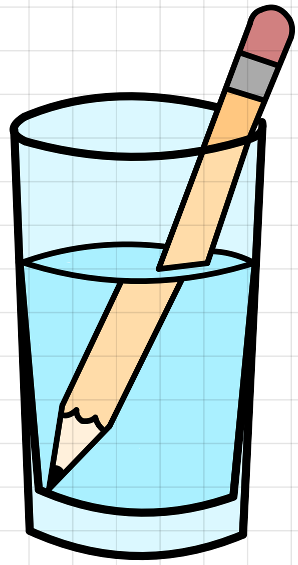


PERISCOPE:

Periscope is an optical device which is normally used in submarines to observe the situation above the water surface when submarine is below the water surface. It works on total internal reflection done by the set of two prisms.



REFRACTION OF LIGHT:



"The bending of light due to change in medium is called refraction of Light"

LAWS OF REFRACTION:

- Incident ray, refracted ray & normal lie on same plane.
- SNELL'S LAW:

"The ratio of Sine of angle of incidence over Sine of angle of refraction is always constant from less dense to denser medium. This constant is known as refractive index (n)."

$$\boxed{n = \frac{\sin\angle i}{\sin\angle r}}$$

Refractive index can also be defined as:

"The ratio of speed of light in vacuum over the speed of Light in medium."

$$\boxed{n = \frac{c}{v}}$$

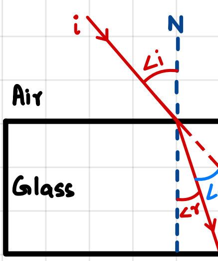
CASES OF REFRACTION:

LESS DENSE → DENSER

Light slows down.

Moves towards normal.

$n = \frac{\sin\angle i}{\sin\angle r}$

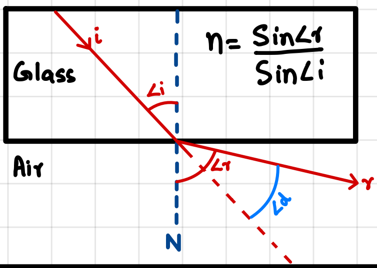


DENSER → LESS DENSE

Light speeds up.

Moves away from normal.

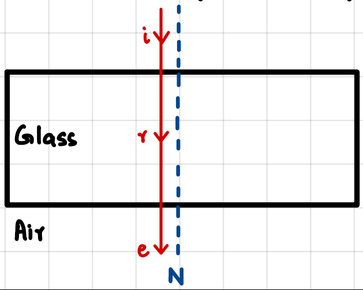
$n = \frac{\sin\angle r}{\sin\angle i}$



PARALLEL TO NORMAL

Light doesn't bend.

Slows down/speeds up.

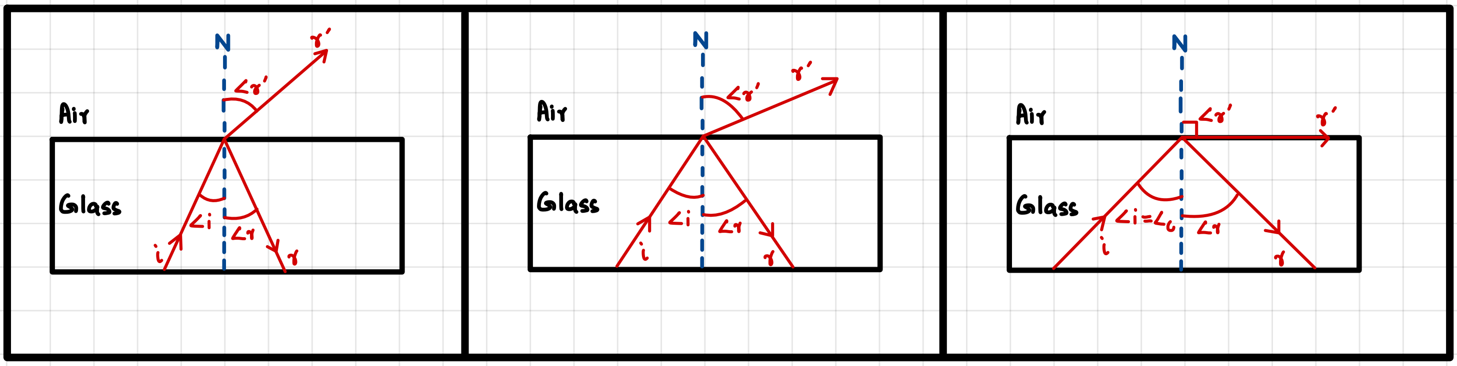


CRITICAL ANGLE:

"Special angle of incidence from denser to less dense at which refracted ray is perpendicular from the normal is called critical angle ($\angle c$)."

Formula of refractive index for critical angle.

$$\boxed{n = \frac{1}{\sin \angle c}}$$

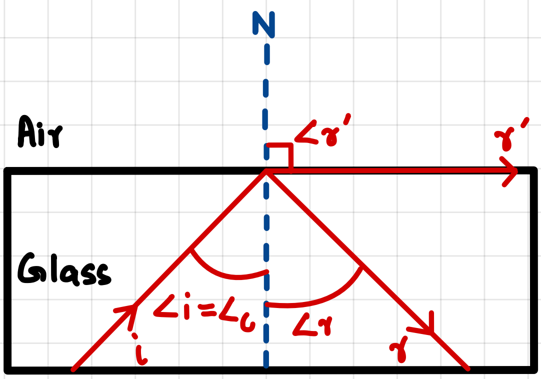


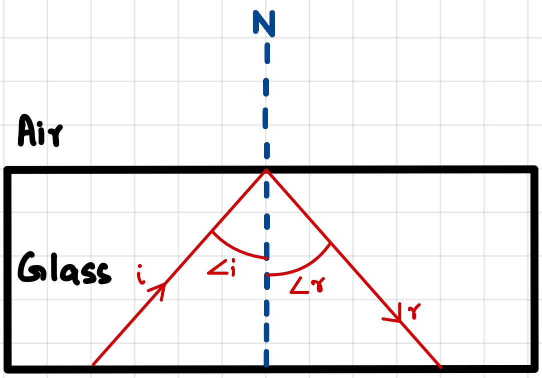
TOTAL INTERNAL REFLECTION:

"The special phenomenon in which light doesn't refract it only reflects is called Total internal reflection."

CONDITIONS:

- Light goes from denser to less dense.
- Angle of incidence must be greater than critical angle ($\angle i > \angle c$).



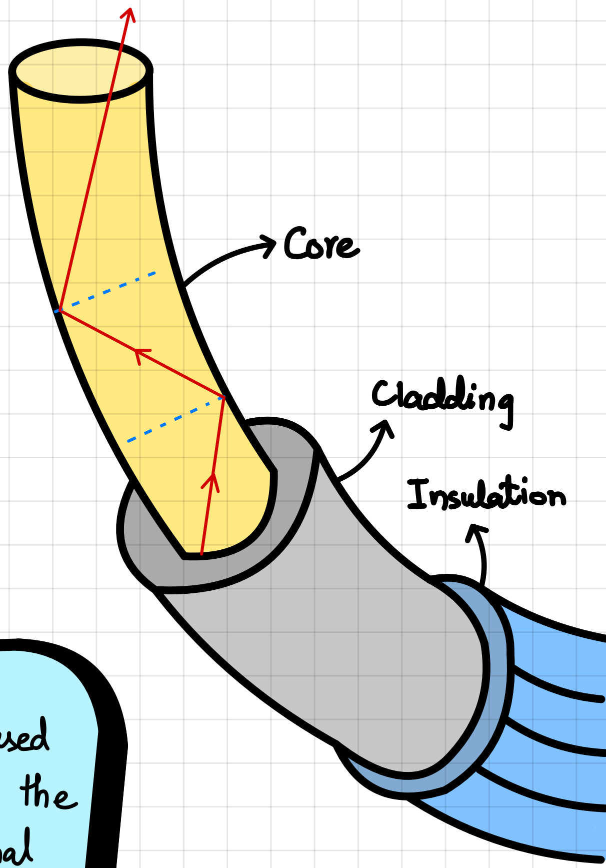


Optical Fibre:

"Wires made from glass which are used for communication are called optical Fibre."

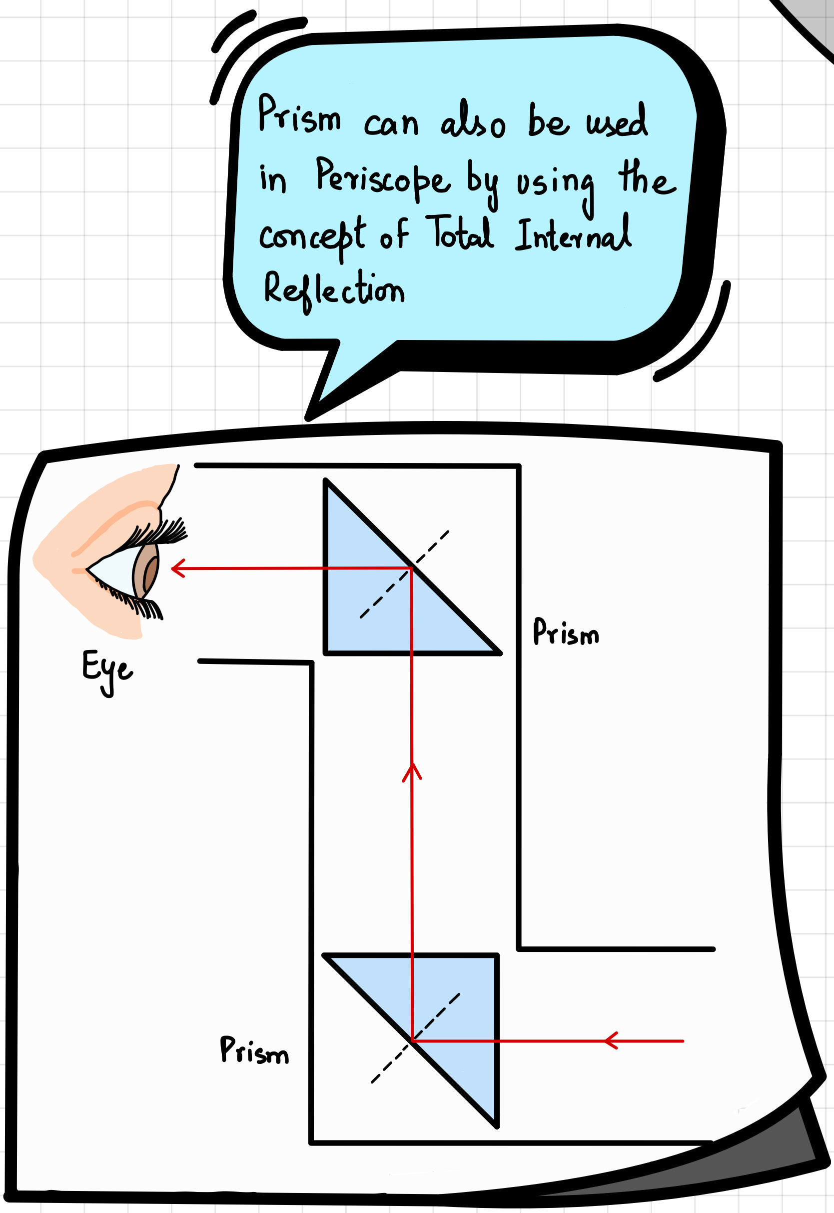
They are made from two type of glasses:

- Core (High refractive index).
- Cladding (Low refractive index).



ADVANTAGES:

- High speed communication.
- Less loss of data.
- Less bulky.
- Cheaper.



3.2.3 Thin Lenses

LENSES:

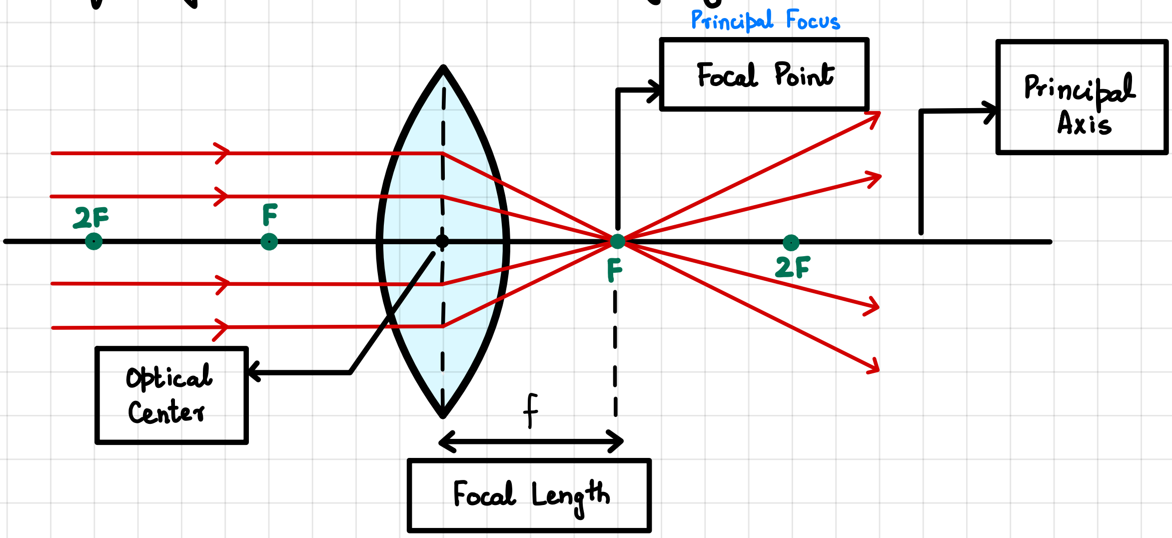
"Thick transparent materials usually made from glass / plastic which are used to converge / diverge rays of light are called Lenses."

CONVERGING (CONVEX) LENS:

"A lens which is thick from center & thin from edges, used to converge rays of light is called converging lens."

SCAN ME

Converging lens burns paper using sunlight



FOCAL POINT:

"The point on Principal Axis at which all rays of light are converged is called Focal Point."

It is also known as Principal Focus.

OPTICAL CENTER:

"Geometrical center of the lens is called optical center."

FOCAL LENGTH:

"The distance between focal point & optical center is called focal length."

It is denoted by 'f'.

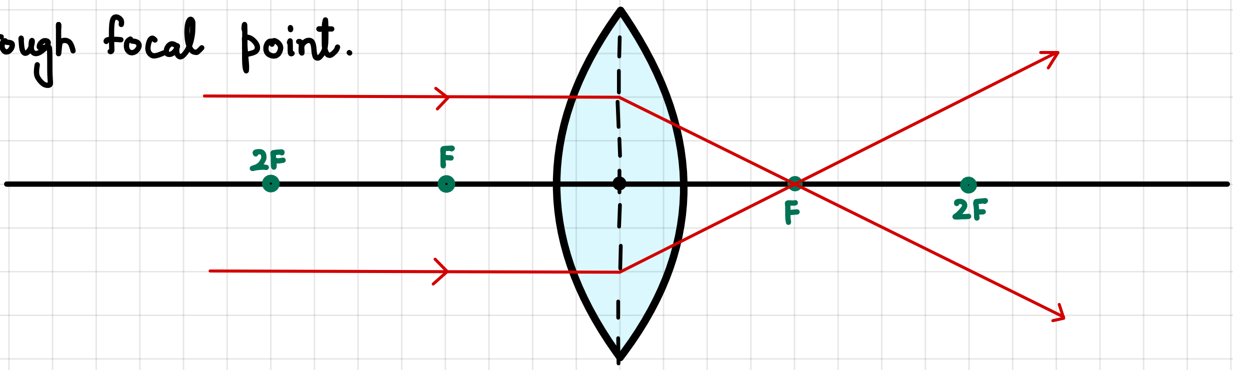
PRINCIPAL AXIS:

"The horizontal line which passes through optical center & focal point is called Principal Axis."

ELEMENTARY RAYS FOR CONVERGING LENS:

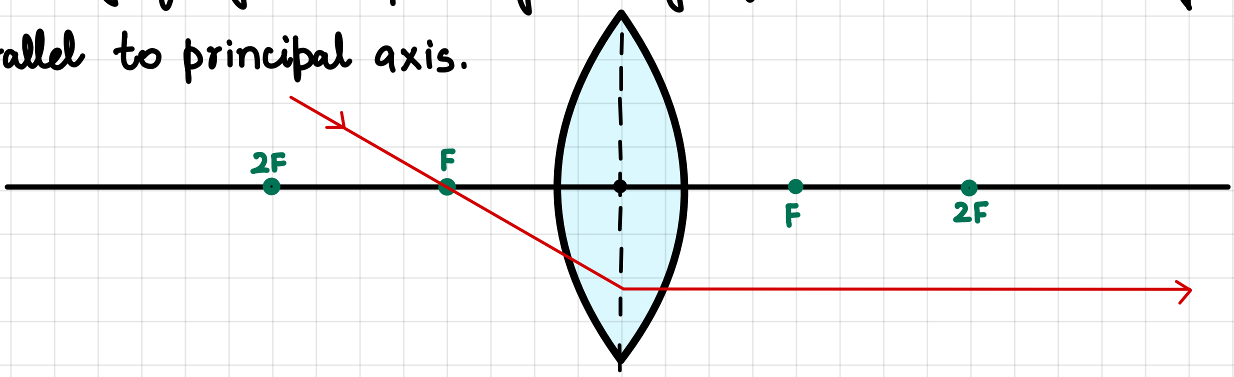
RAY # 01:

When ray of light is parallel to the principal axis, it alway goes through focal point.



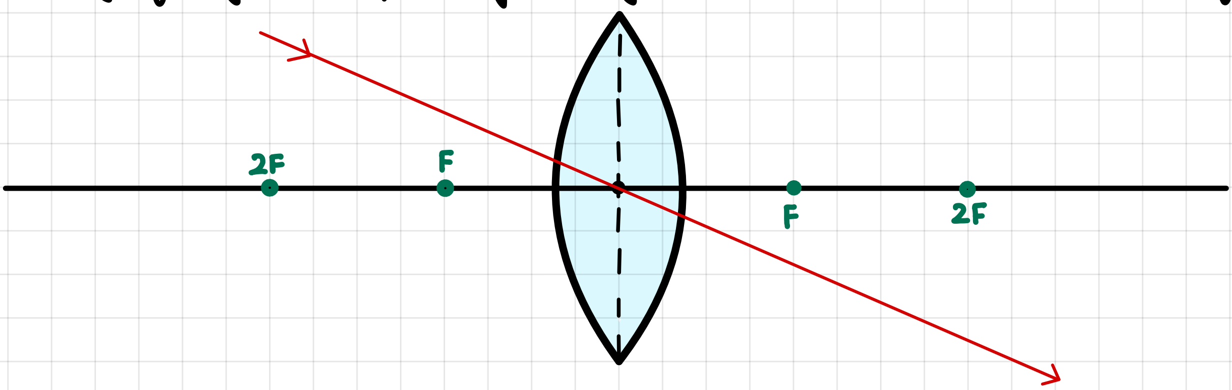
RAY # 02:

When ray of light is passing through focal point it always goes parallel to principal axis.



RAY # 03:

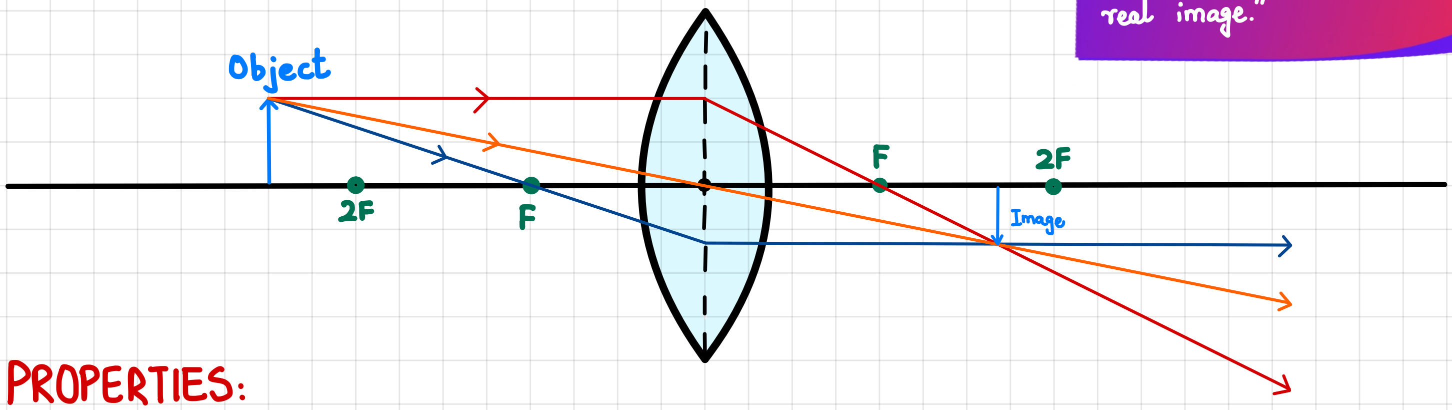
When ray of light is passing through the center, it passes undeflected.



Ray Diagrams:

① WHEN OBJECT IS PLACED BEYOND 2F

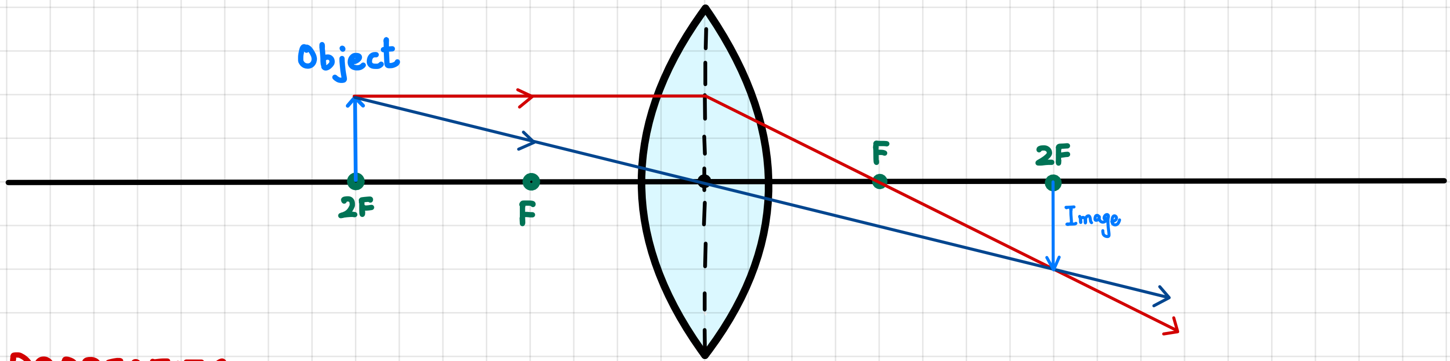
Real Image.
"An image which forms on a screen is called real image."



PROPERTIES:

- Real Image
- Inverted Image
- Diminished Image
- Used in Camera

② WHEN OBJECT IS PLACED AT 2F



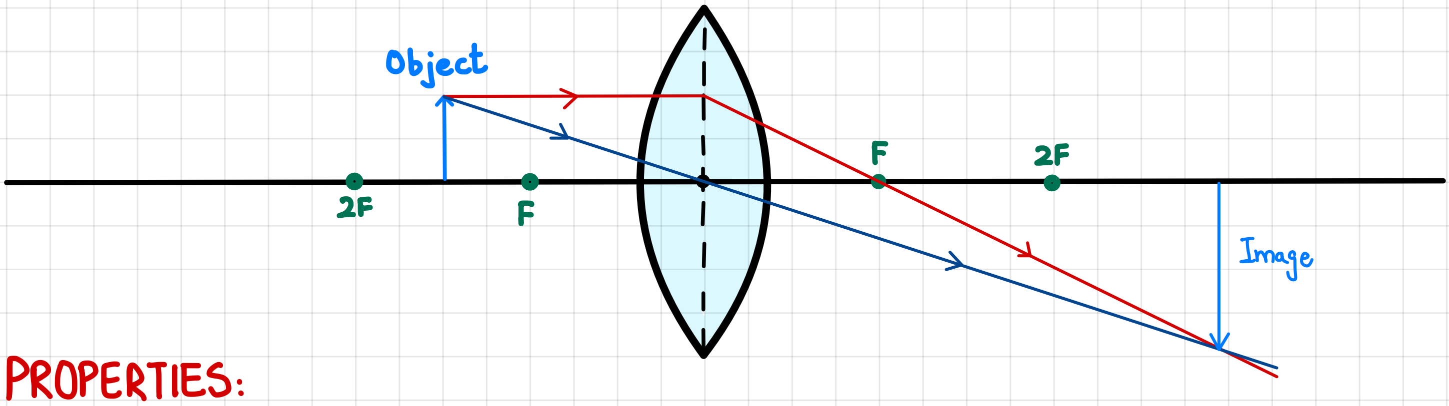
PROPERTIES:

- Real Image
- Inverted Image
- Same size Image
- Used in Photocopier

SCAN ME

Image formed by Converging Lens

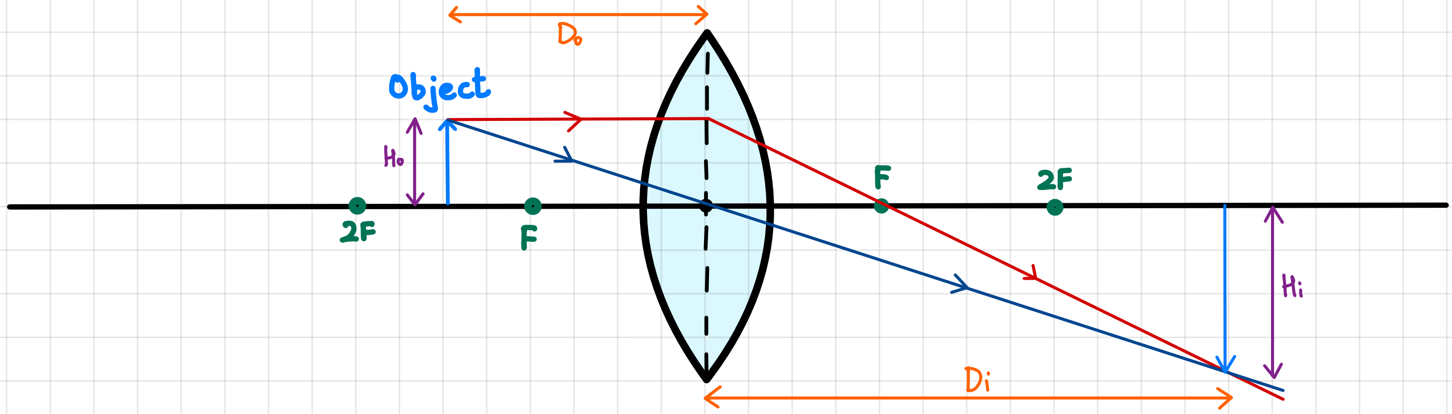
③ WHEN OBJECT IS PLACED BETWEEN F & 2F



PROPERTIES:

- Real Image
- Inverted Image
- Magnified Image
- Used in Projector

LINEAR MAGNIFICATION:



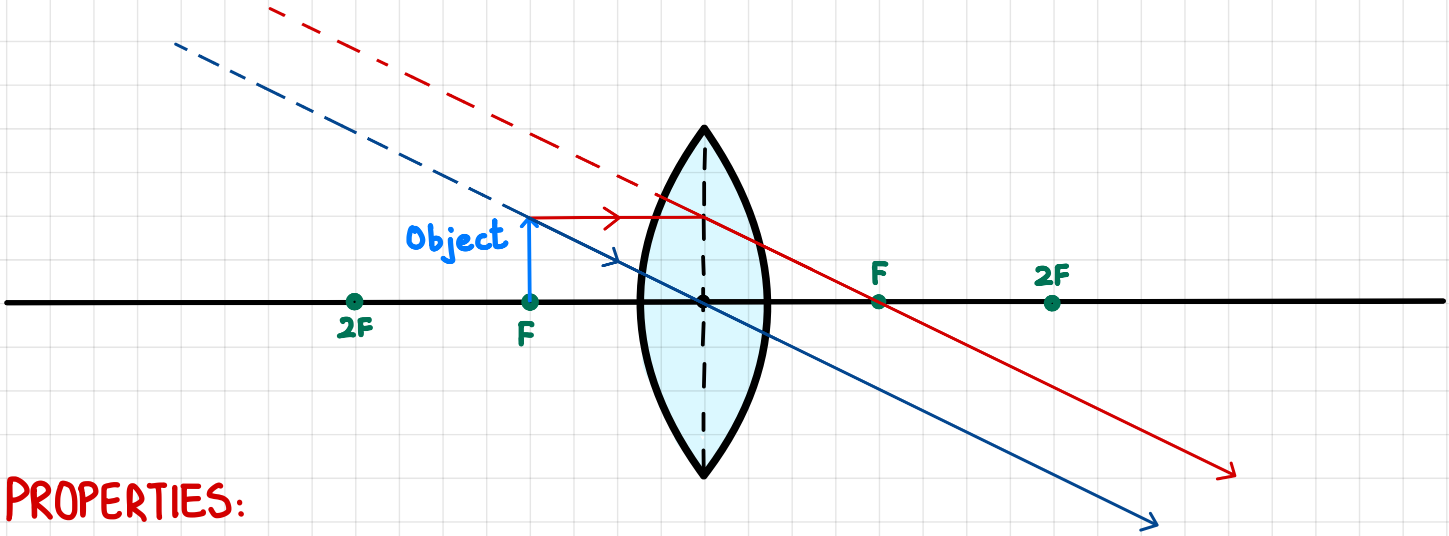
"The ratio of image height over object height is called Linear magnification."

$$\boxed{M = \frac{H_i}{H_o}}$$

"The ratio of image distance over object distance is called Linear magnification."

$$\boxed{M = \frac{D_i}{D_o}}$$

④ WHEN OBJECT IS PLACED AT F:

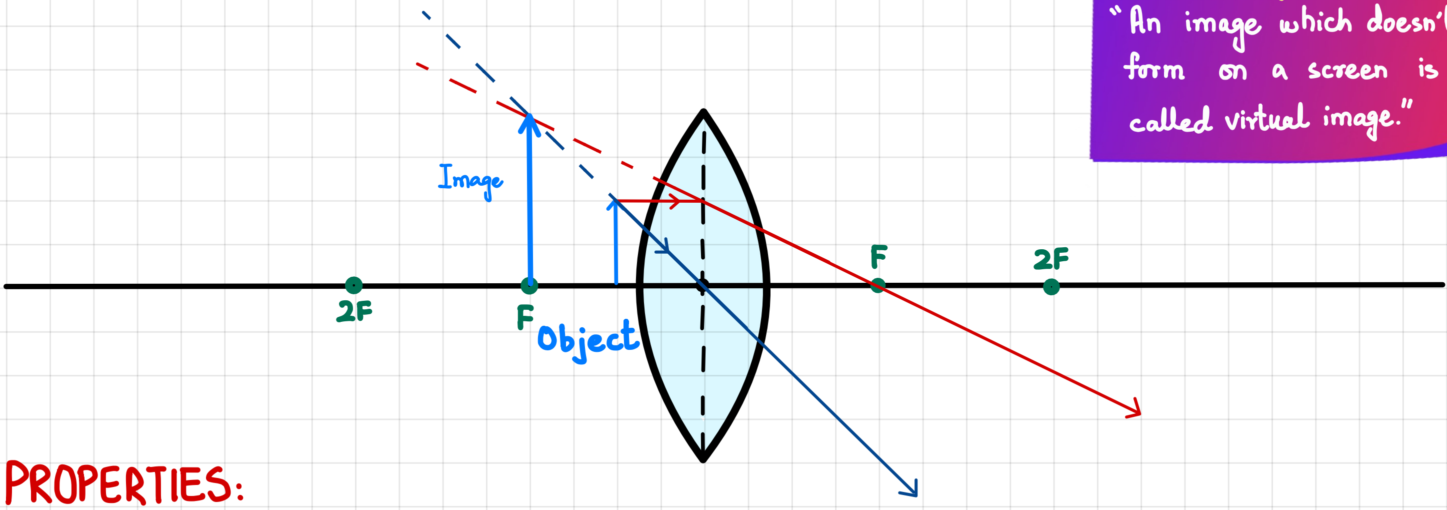


PROPERTIES:

- Virtual Image
- Magnified Image
- Upright Image
- Used in Sky beam

⑤ WHEN OBJECT IS PLACED BETWEEN F & O

Virtual Image:
"An image which doesn't form on a screen is called virtual image."

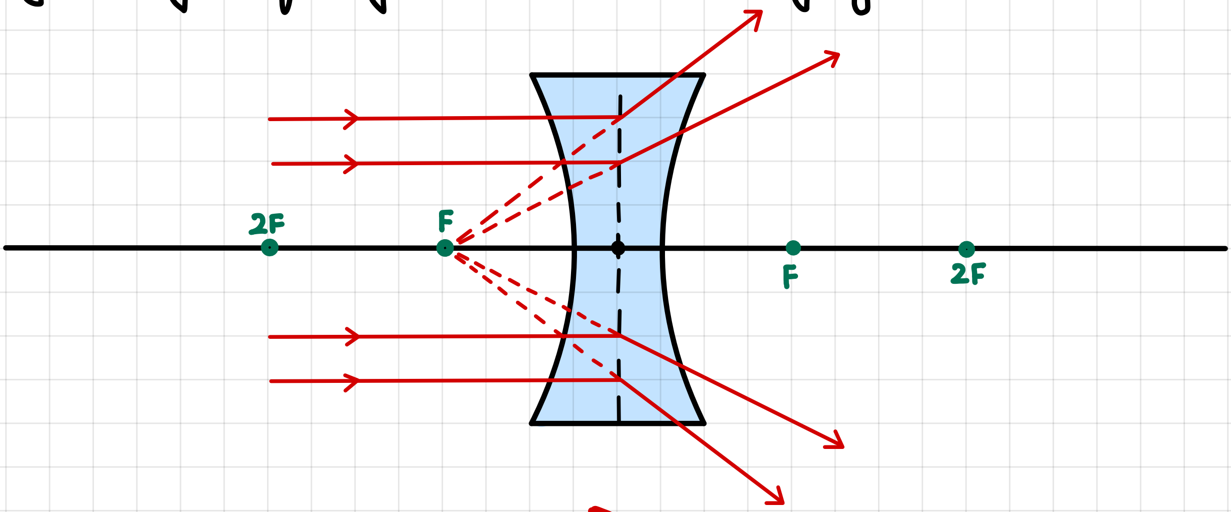


PROPERTIES:

- Virtual Image
- Magnified Image
- Upright Image
- Used in Magnifying glass

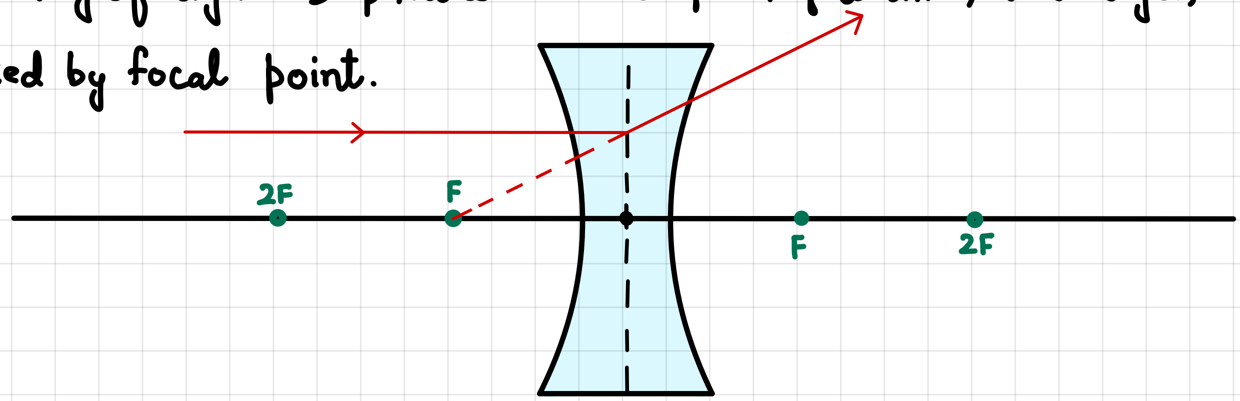
DIVERGING (CONCAVE) LENS:

"A lens which is thin from center & thick from edges, used to diverge rays of light is called diverging lens."

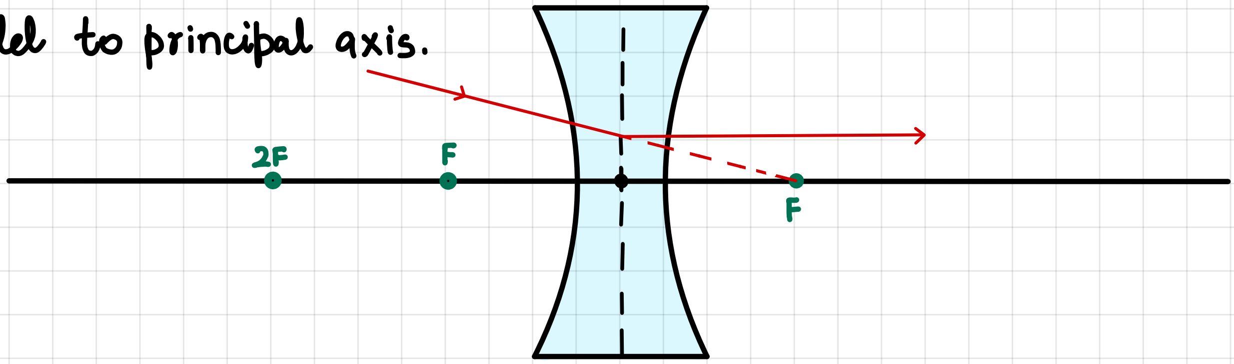


ELEMENTARY RAYS FOR DIVERGING LENS:

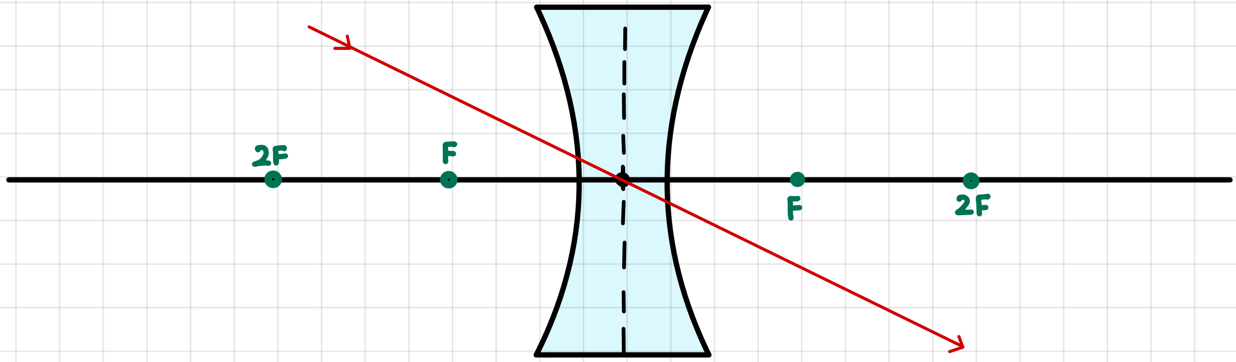
RAY # 01:
When ray of light is parallel to the principal axis, it diverges, directed by focal point.



RAY # 02:
When ray of light is directed to the focal point it always goes parallel to principal axis.

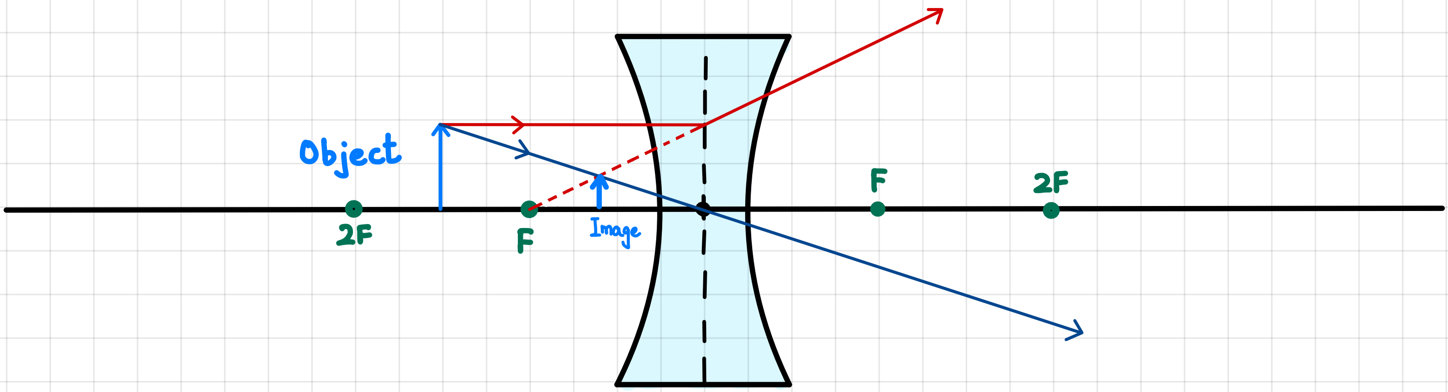


RAY # 03:
When ray of light is passing through the center, it passes undeflected.



Ray Diagrams:

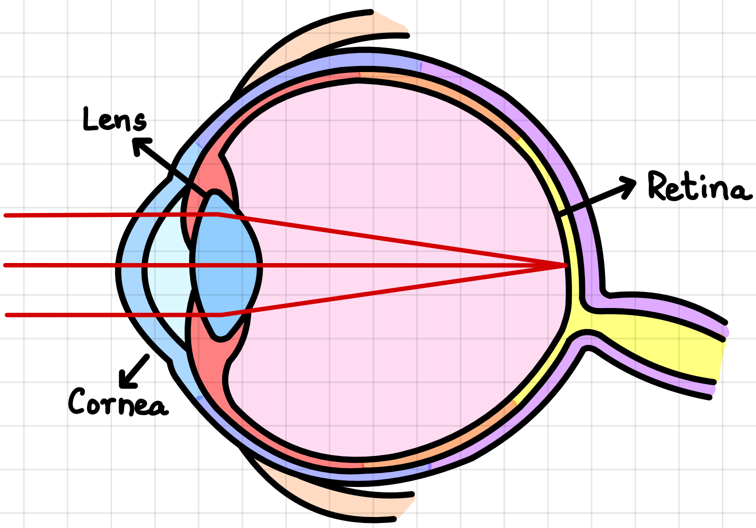
③ WHEN OBJECT IS PLACED BETWEEN F & 2F



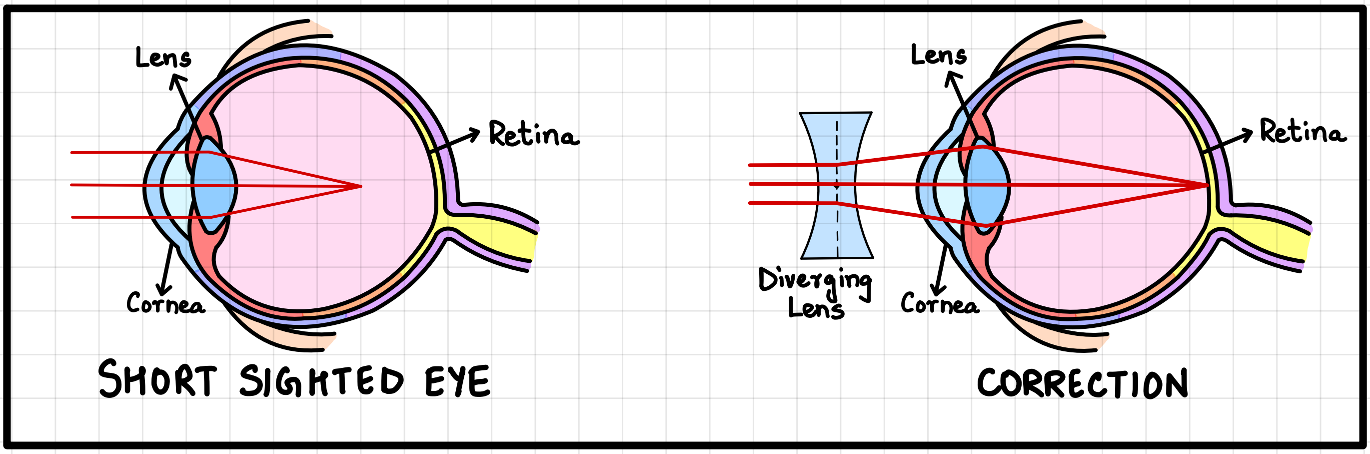
PROPERTIES:

- Virtual Image
- Diminished Image
- Upright Image
- Used in short sighted glasses.

Normal Human Eye:



SHORT SIGHTEDNESS:

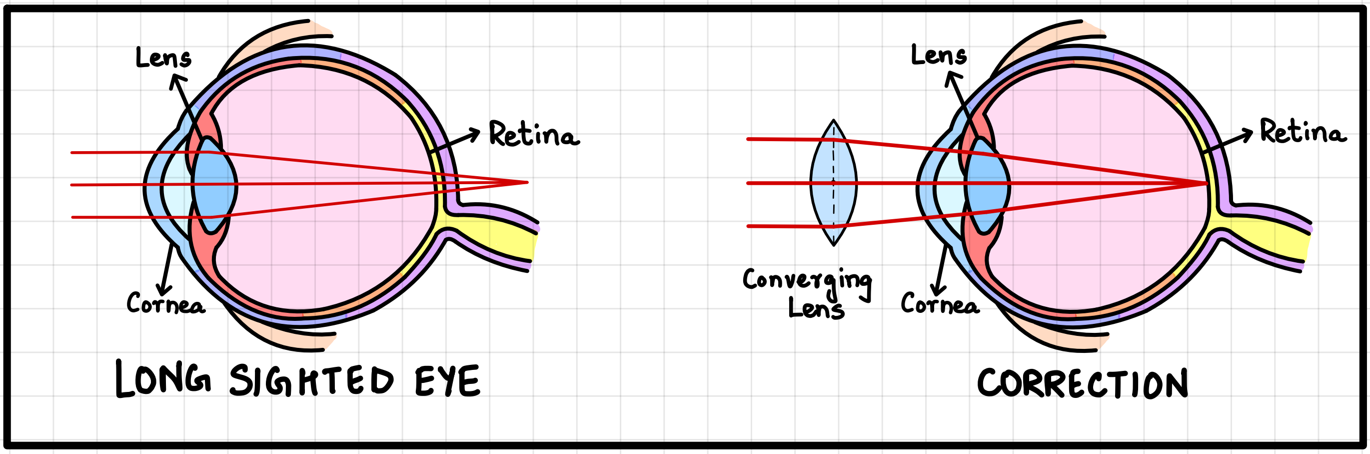


"A disease in which a person can see near objects clearly but can't see distant objects clearly is called short sightedness"

In this disease, an eye ball expands, lens remains same & image forms before retina.

- It can be corrected using diverging Lens.

LONG SIGHTEDNESS:



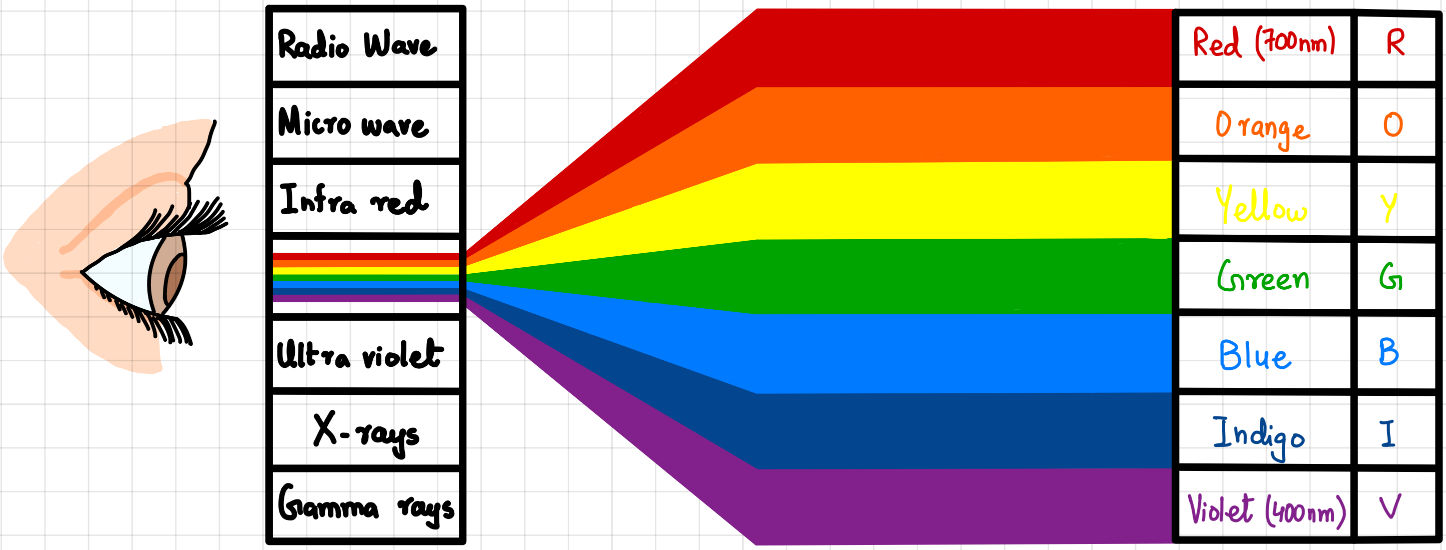
"A disease in which a person can see distant objects clearly but can't see near objects clearly is called long sightedness"

In this disease, an eye ball contracts, lens remains same & image forms beyond retina.

- It can be corrected using converging Lens.

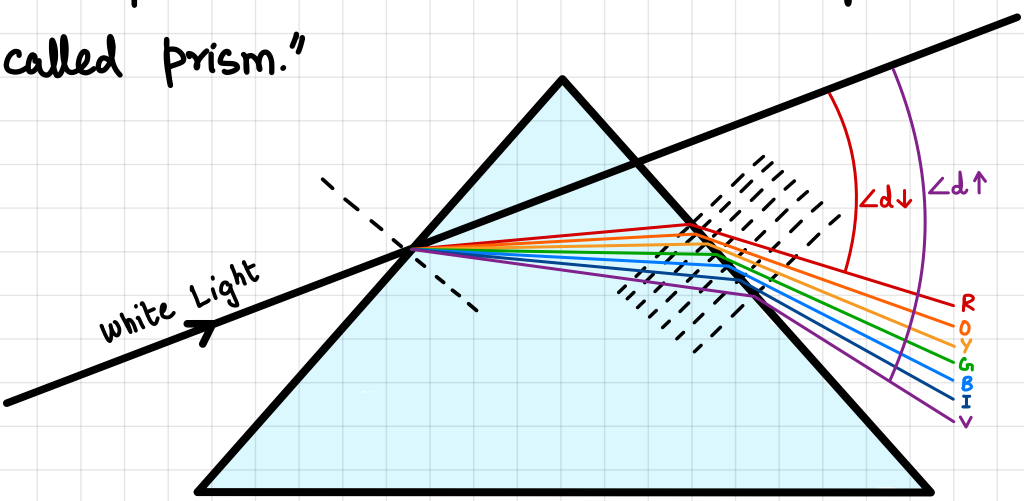
Visible Spectrum:

We can only see lights of wavelength between 400 – 700 nm. This is called visible spectrum which consists of seven colours namely red, orange, yellow, green, blue, indigo & violet. Wavelengths less than 400 nm or greater than 700 nm can't be seen by normal human eye.



Prism:

"An optical instrument used to disperse rays of light is called prism."



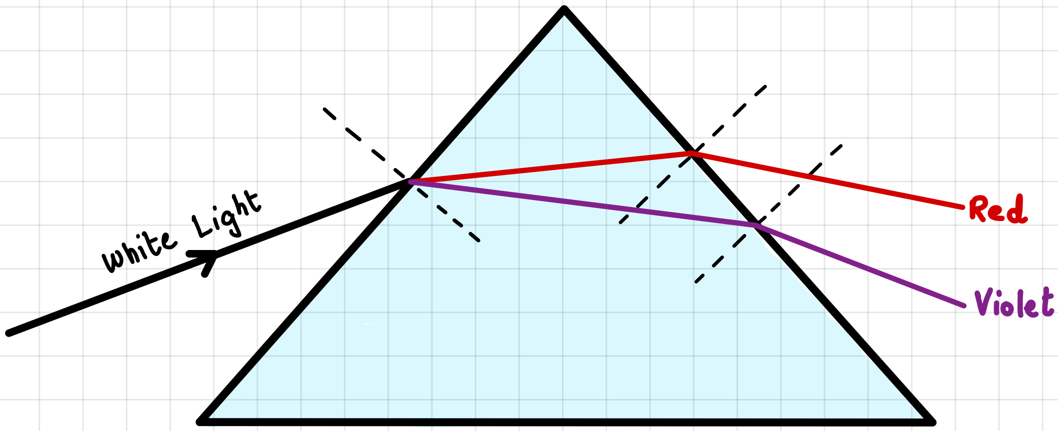
SCAN ME

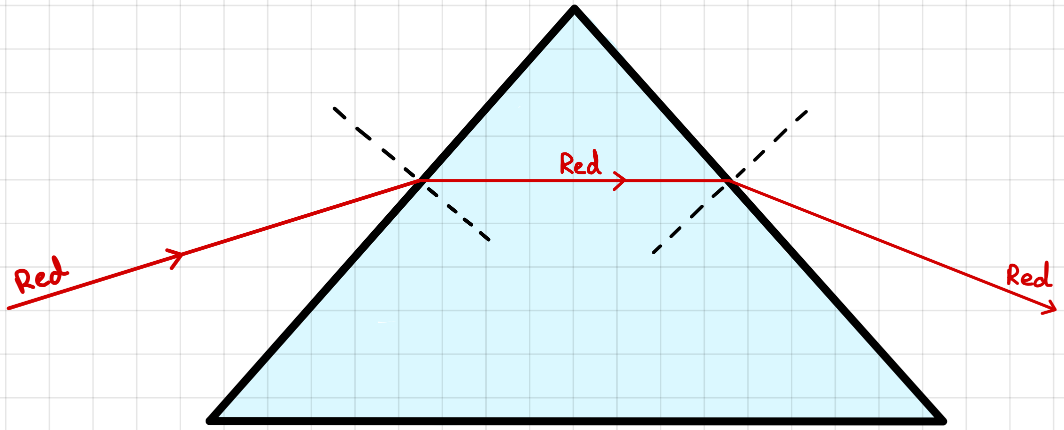
Dispersion of Sun Light by Prism

- Deviates most: Violet
- Deviates least: Red

Splitting of white light into seven different colours is called Dispersion of Light.

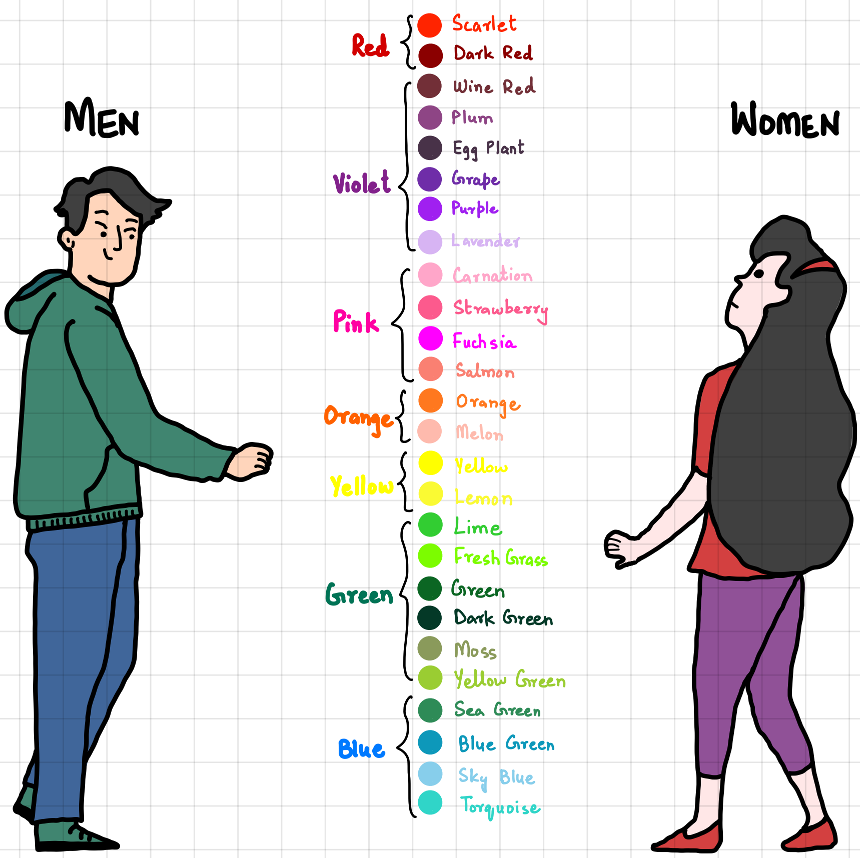
DRAW THE PATTERN OF DISPERSION (RED & VIOLET ONLY):





Prism doesn't disperse single colour light.

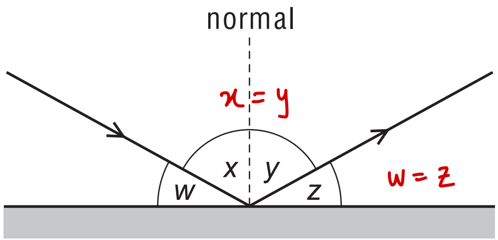
How We See Colours



MATCH THE FOLLOWING

In reflection,	imaginary line which makes 90° with the surface.
Normal is an	refractive index is calculated by $n = \frac{\sin\angle r}{\sin\angle i}$.
Refraction defines,	angle of incidence = angle of reflection.
When light moves dense to less dense	the bending of light as it enters from dense to less dense medium or vice versa.
When light slows down,	when $\angle i > \angle c$ from denser to less dense.
Critical angle defines,	it moves towards the normal.
Total internal reflection happens	Virtual images.
Focal length is	is upright, virtual & magnified.
Diverging Lens forms	special angle of incidence from dense to less dense at which refracted ray is perpendicular from the normal.
Image formed by magnifying glass	image forms in front of retina & is corrected by diverging lens.
In short-sightedness,	the distance b/w optical center & focal point.

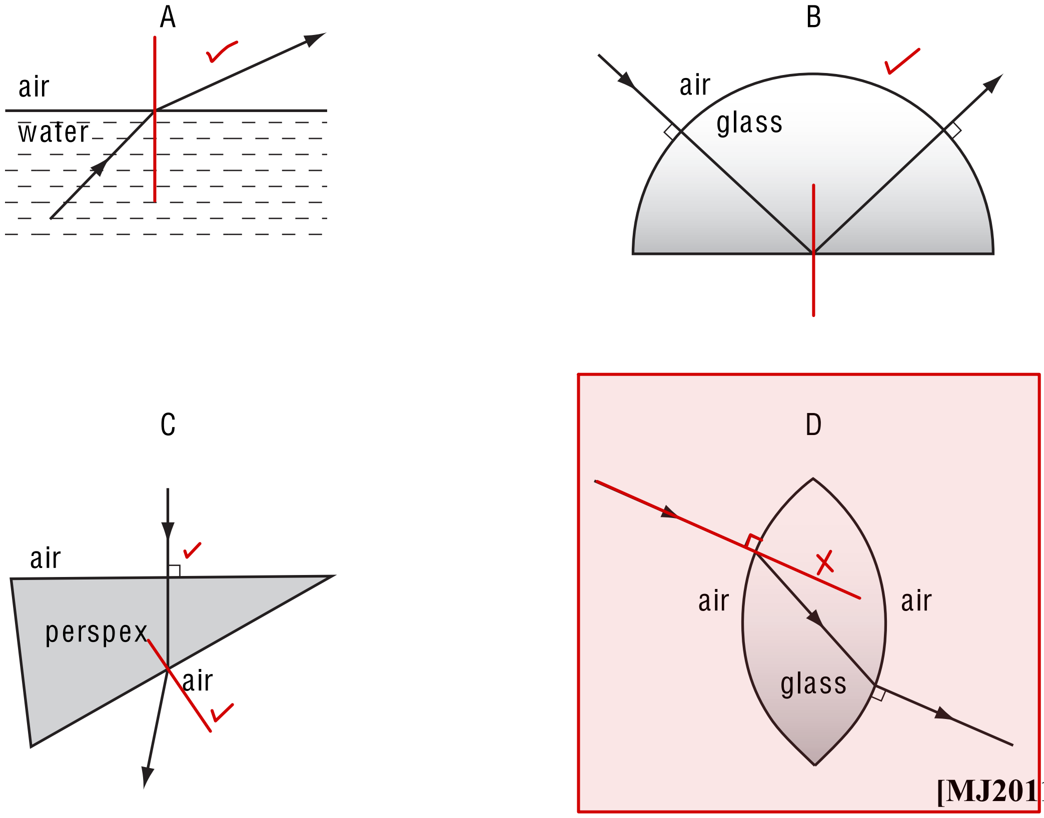
01 A ray of light strikes a plane mirror and is reflected.



Which pair of angles **must** be equal in value?

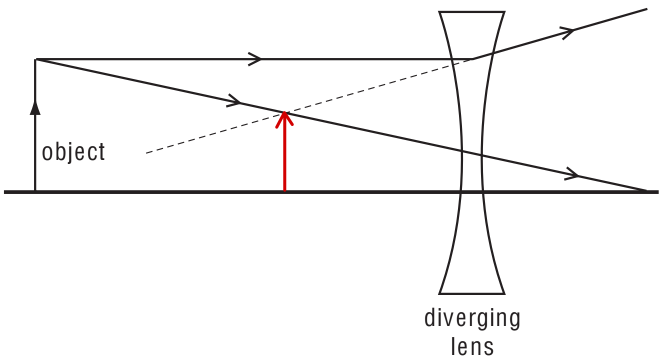
A *w* and *x* **B** *w* and *y* **C** *x* and *y* **D** *x* and *z* **[MJ2011/P11/Q21]**

02 In which diagram is the path of the light ray **not** correct?



[MJ2011/P11/Q22]

03 The ray diagram shows two rays from a point on an object placed in front of a diverging (concave) lens.



What are the properties of the image produced?

A real and larger than the object

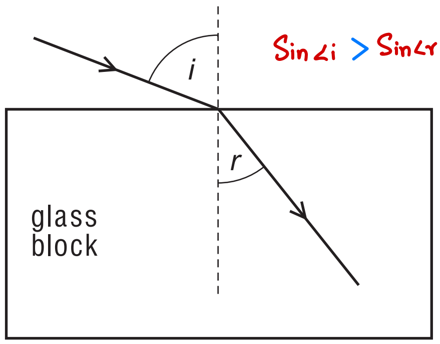
B real and smaller than the object

C virtual and larger than the object

D virtual and smaller than the object

[MJ2011/P11/Q23]

04 A ray of light enters a glass block at an angle of incidence *i*, producing an angle of refraction *r* in the glass.



$\sin\angle i > \sin\angle r$

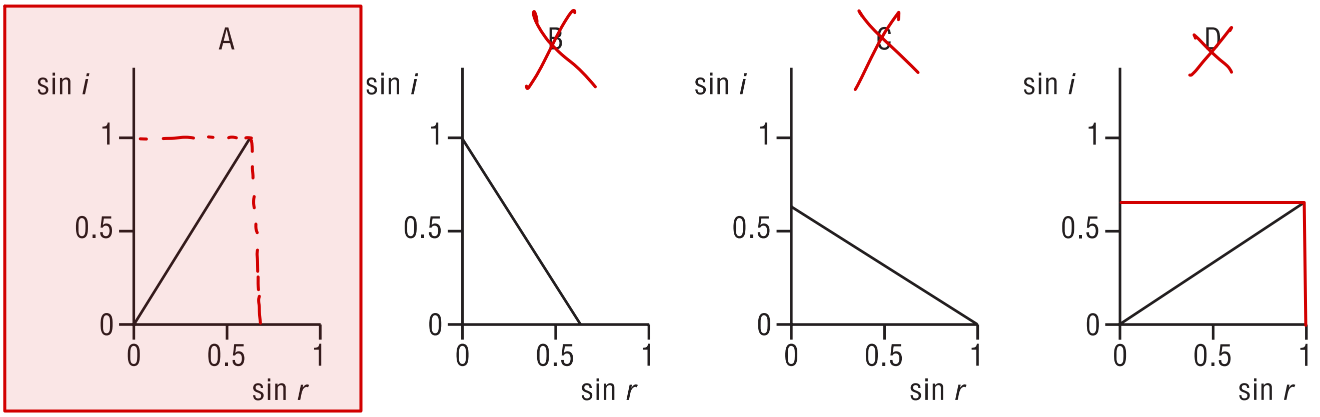
$n = \dfrac{\sin\angle i}{\sin\angle r}$

$\sin\angle i = n \sin\angle r$

$\sin\angle i \propto \sin\angle r$

Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

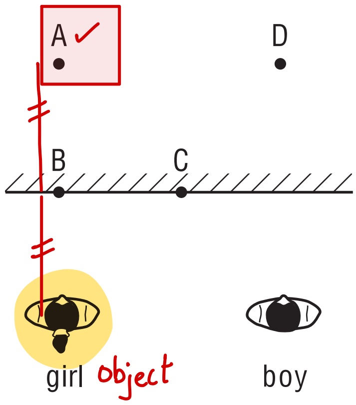
Which graph is correct?



[ON2011/P11/Q21]

05 A boy stands beside a girl in front of a large plane mirror. They are both the same distance from the mirror, as shown.

Where does the boy see the girl's image?



[ON2011/P11/Q22]

06 Light is incident on one face of a glass block at an angle of incidence of 40°. The glass block is in air.

Air → Glass

The refractive index of the glass is 1.46.

$n = \dfrac{\sin\angle i}{\sin\angle r}$, $\sin\angle r = \dfrac{\sin\angle i}{n}$

What is the angle of refraction inside the glass block?

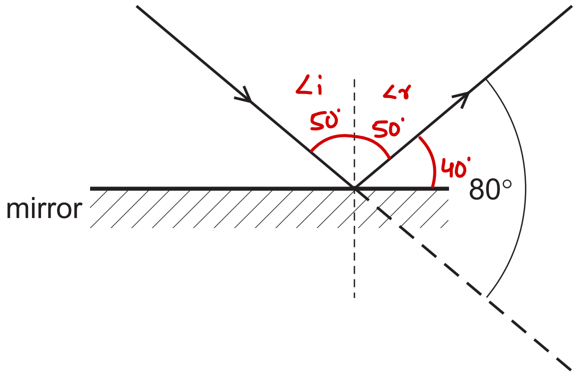
$\angle r = \sin^{-1}\left(\dfrac{\sin 40^\circ}{1.46}\right)$

A 26° **B** 27° **C** 58° **D** 70°

[MJ2012/P11/Q23]

$\angle r = 26.12^\circ$

07 Light is incident on a mirror and is reflected as shown.



What is the angle of incidence and the angle of reflection?

	angle of incidence / °	angle of reflection / °
A	40	40
B	40	50 ✓
C	50 ✓	40
D	50 ✓	50 ✓

[MJ2012/P11/Q22]

08 In a short-sighted eye, light from distant objects is not focused on the retina.

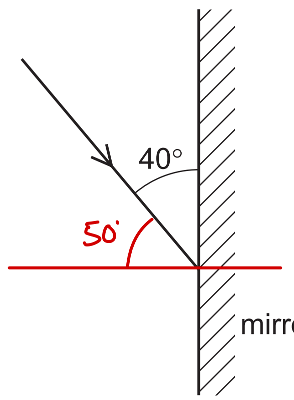
Where is this light focused and what type of lens is needed to correct the problem?

	where focused	lens needed
A	behind the retina	converging lens
B	behind the retina	diverging lens ✓
C	in front of the retina ✓	converging lens
D	in front of the retina ✓	diverging lens ✓

[MJ2012/P11/Q24]

09 The diagram shows a ray of light directed at a plane mirror.
What are the angle of incidence and the angle of reflection?

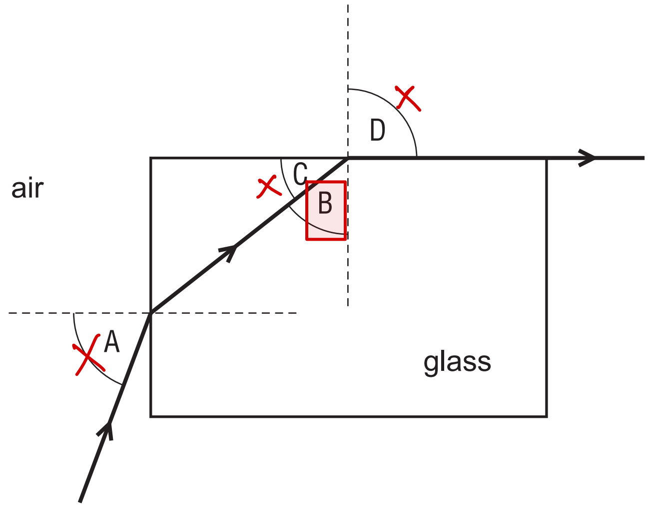
	angle of incidence	angle of reflection
A	40°	40°
B	40°	50° ✓
C	50° ✓	40°
D	50° ✓	50° ✓



[MJ2012/P12/Q22]

10 Light travels through a glass block as shown.

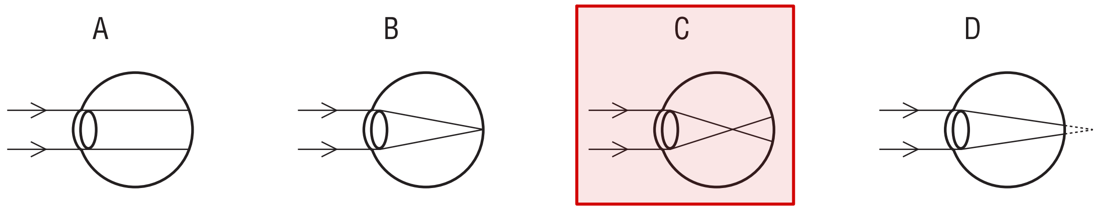
Which angle is the critical angle for light in the glass?



[MJ2012/P12/Q23]

11 A man is short-sighted.

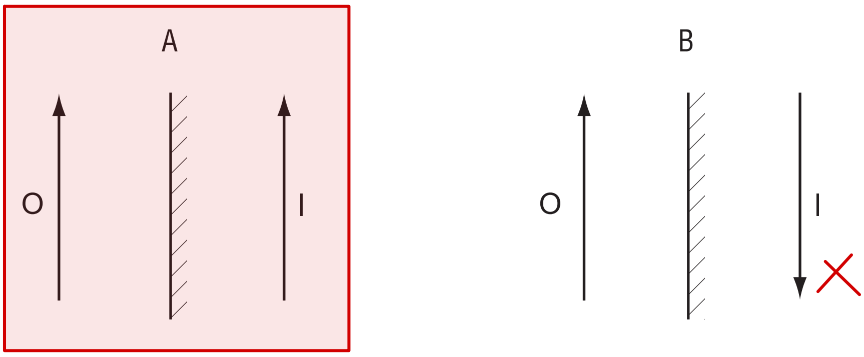
Which ray diagram shows what happens in his eye when he looks at a distant object?

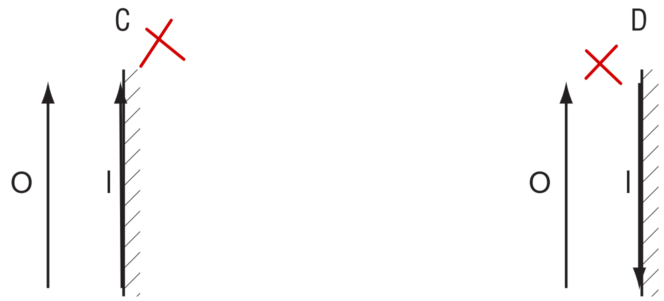


[MJ2012/P12/Q24]

12 An object O is placed in front of a plane mirror.

Which diagram correctly represents the image I formed by the mirror?

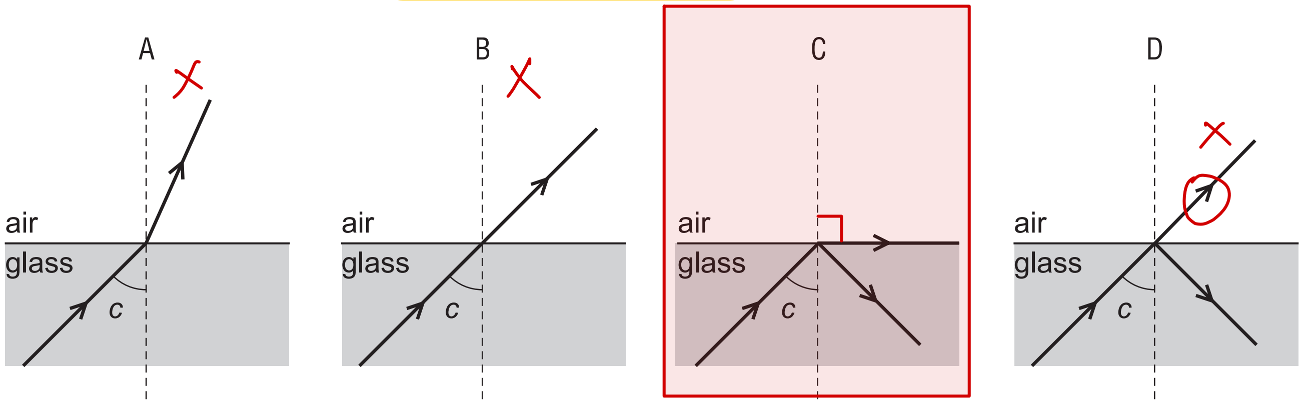




[ON2012/P11/Q22]

13 In the following diagrams, the angle c is the critical angle.

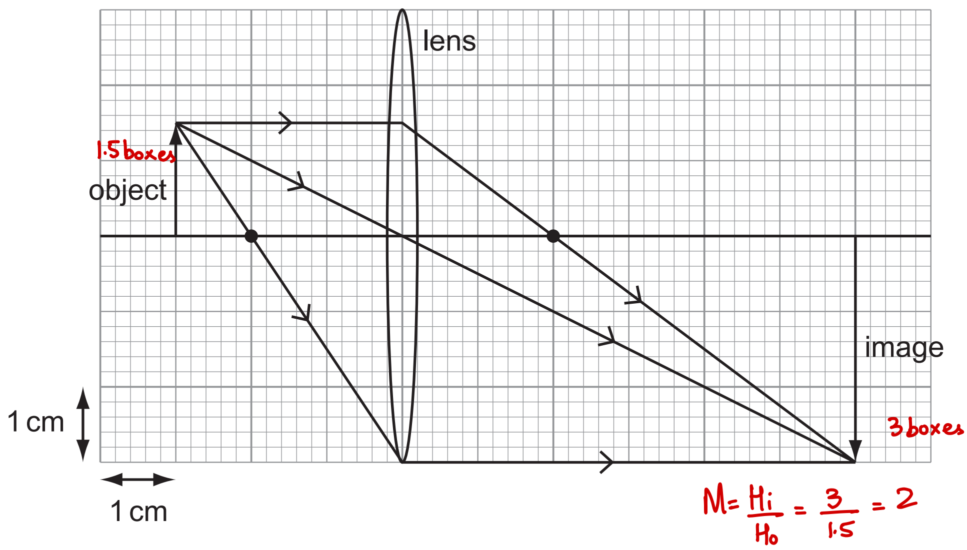
Which diagram shows the correct path of the light ray?



[ON2012/P11/Q23]

14 An object of height 1.5 cm is placed in front of a converging lens of focal length 2.0 cm.

The arrangement is shown on the full-scale ray diagram.



$M = \frac{H_i}{H_o} = \frac{3}{1.5} = 2$

What is the linear magnification produced by the lens?

A 2.0 **B** 3.0 **C** 4.0 **D** 6.0 **[ON2012/P11/Q24]**

15 An object is placed in front of a plane mirror. The image produced is

A real and smaller than the object.

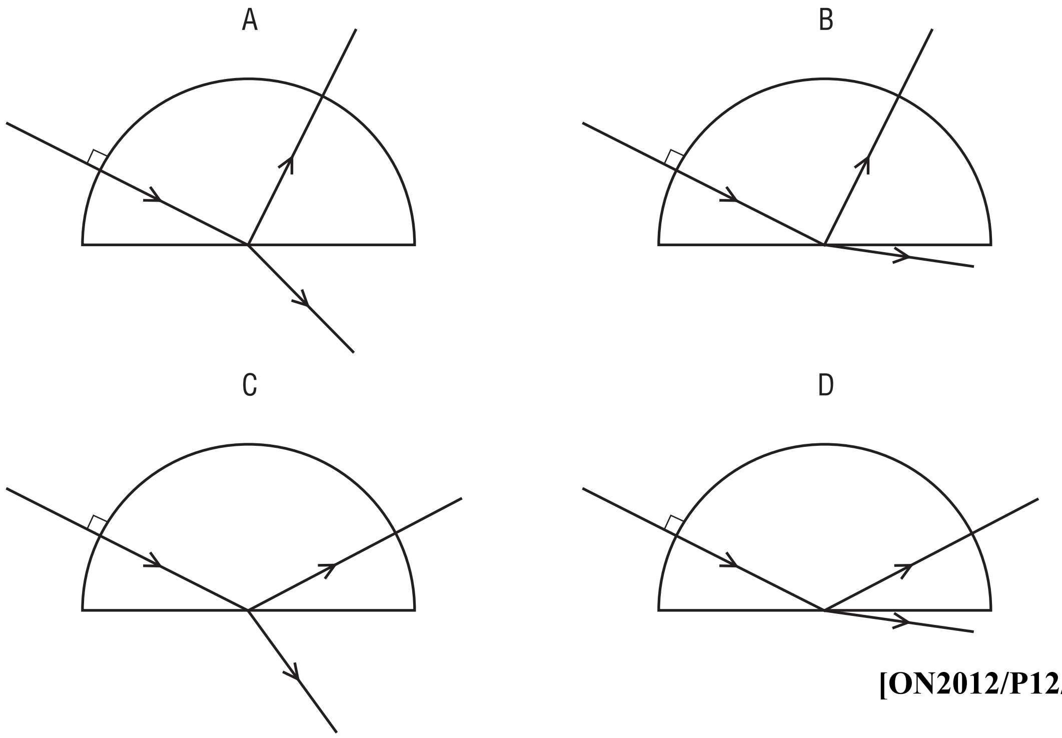
B real and the same size as the object.

C virtual and smaller than the object.

D virtual and the same size as the object. **[ON2012/P12/Q22]**

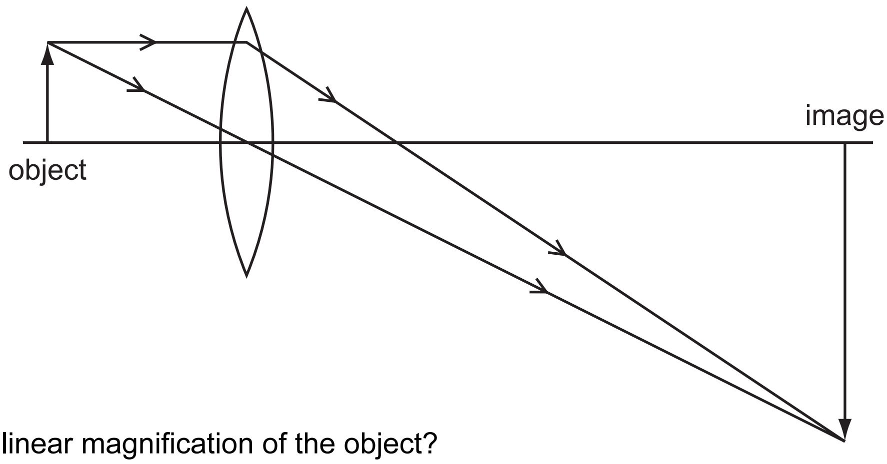
16 A ray of red light enters a semi-circular glass block normal to the curved surface.

Which diagram shows the partial reflection and refraction of the ray?



[ON2012/P12/Q23]

17 A lens is used to produce a magnified image, as shown in the scale diagram.



What is the linear magnification of the object?

A 0.33 **B** 3.0 **C** 4.0 **D** 6.0 **[ON2012/P12/Q24]**

18 Which characteristics describe an image formed by a vertical plane mirror?

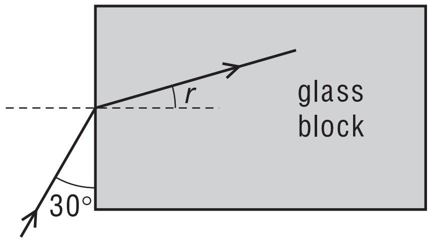
A real and inverted

B virtual and not inverted

C real and larger than the object

D virtual and smaller than the object **[MJ2013/P11/Q19]**

19 A ray of light meets the face of a glass block at an angle of 30° as shown.



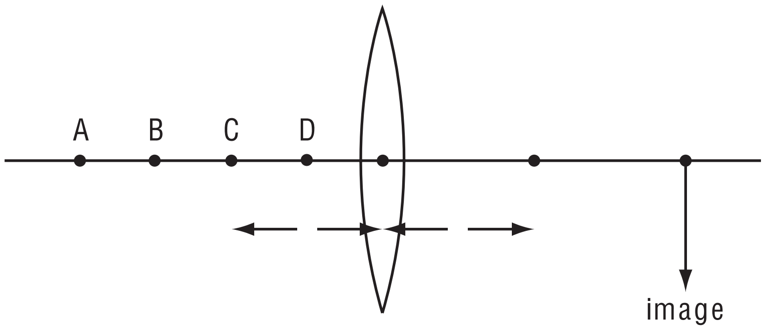
The refractive index of the glass is 1.5.

What is the angle of refraction *r* inside the glass block?

A 19° **B** 20° **C** 35° **D** 40° **[MJ2013/P11/Q20]**

20 The diagram shows a thin converging lens of focal length *f*.

Where must an object be placed to produce a real image in the position shown?



[MJ2013/P11/Q21]

21 Which characteristics describe an image formed by a vertical plane mirror?

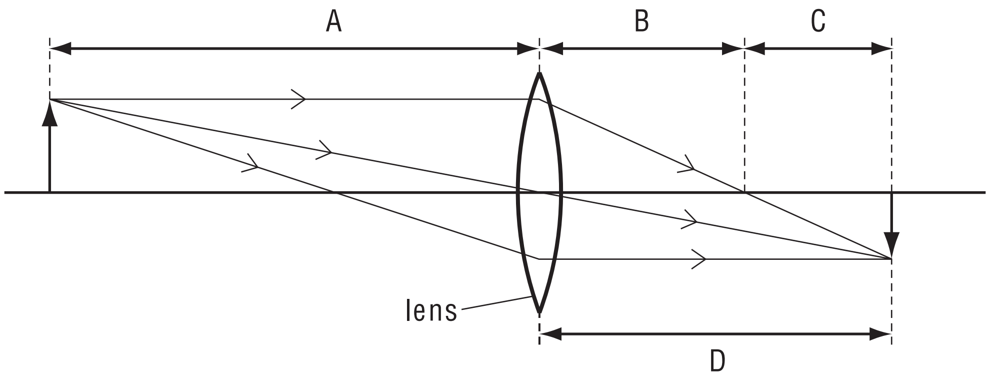
A real and inverted

B virtual and not inverted

C real and larger than the object

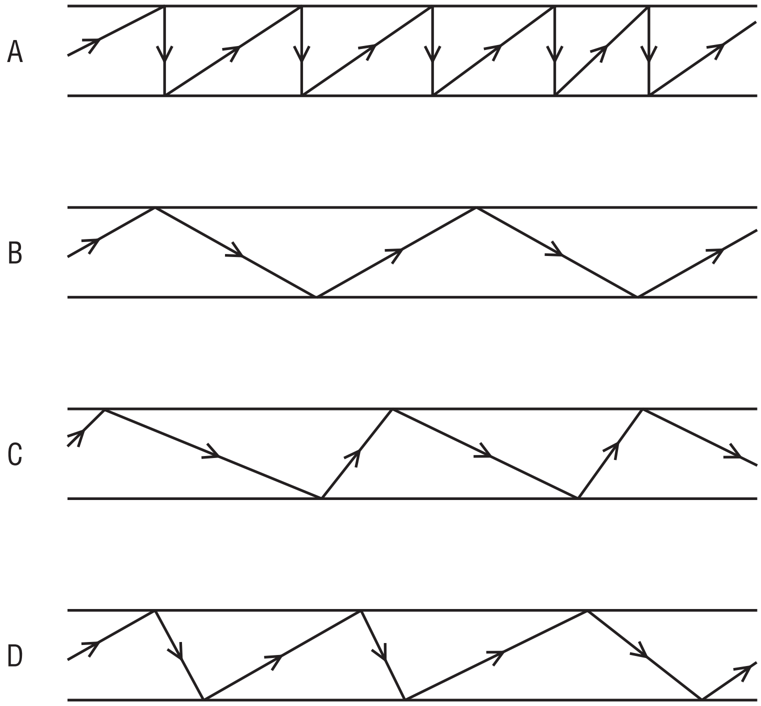
D virtual and smaller than the object **[MJ2013/P12/Q25]**

22 Which length is the focal length of the lens?



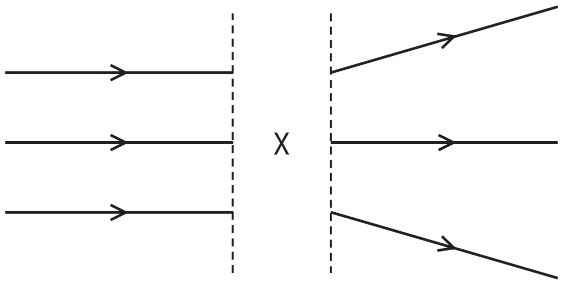
[MJ2013/P12/Q26]

23 Which diagram represents the reflection of light along an optical fibre?



[ON2013/P11/Q20]

24 The diagram shows rays of light.



What is in the space labelled X?

A a converging lens

B a diverging lens

C a plane mirror

D a rectangular glass block

[ON2013/P11/Q21]

25 A ray of red light from a laser passes into a semi-circular glass block.

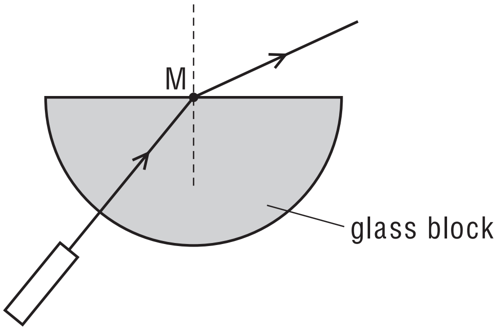
What is shown at M?

A dispersion

B rarefaction

C reflection

D refraction



[ON2013/P12/Q23]

26 A ray of light strikes the surface of a glass block at an angle of incidence of 45°.

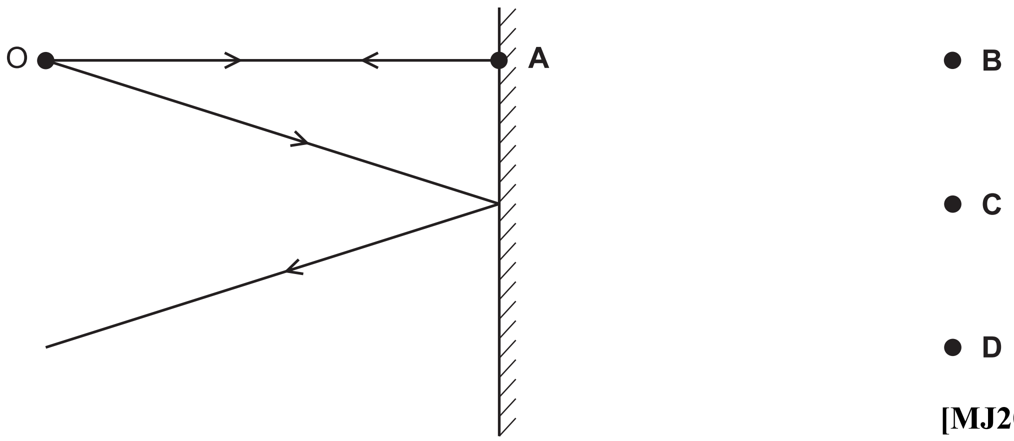
The refractive index of the glass is 1.8.

What is the angle of refraction inside the block?

A 23° **B** 25° **C** 45° **D** 81° **[ON2013/P12/Q24]**

27 The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

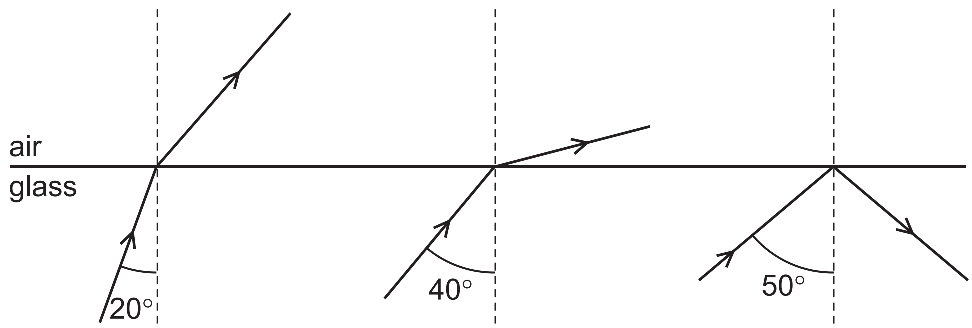
At which position is the image formed?



[MJ2014/P11/Q24]

28 Three rays of light are incident on the boundary between a glass block and air.

The angles of incidence are different.



What is a possible critical angle for light in the glass?

A 15° **B** 30° **C** 45° **D** 60° **[MJ2014/P11/Q25]**

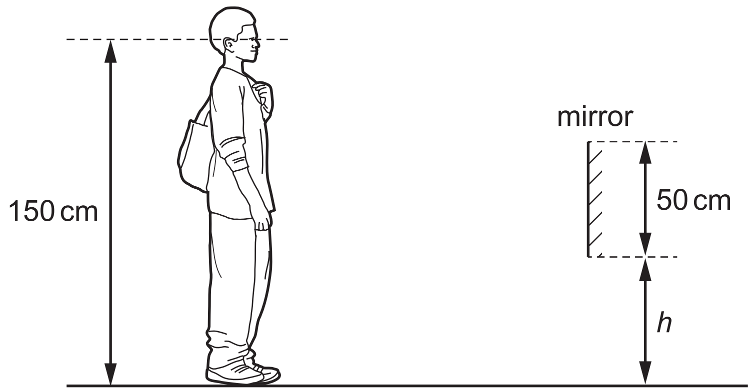
29 Which row applies to a short-sighted eye viewing a distant object?

	position of the image	lens needed for correction
A	behind the retina	converging lens
B	behind the retina	diverging lens
C	in front of the retina	converging lens
D	in front of the retina	diverging lens

[MJ2014/P11/Q27]

30 A shoe shop puts a mirror on the wall so that customers can look at their shoes.

The length of the mirror is 50 cm. A customer has eyes 150 cm above ground level.

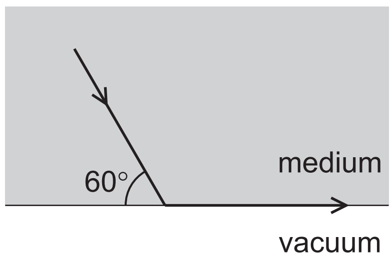


The bottom of the mirror is at height h above the ground.

What is the smallest value of h that allows the customer to see an image of his shoes in the mirror?

A 0 **B** 25 cm **C** 50 cm **D** 75 cm **[ON2014/P11/Q20]**

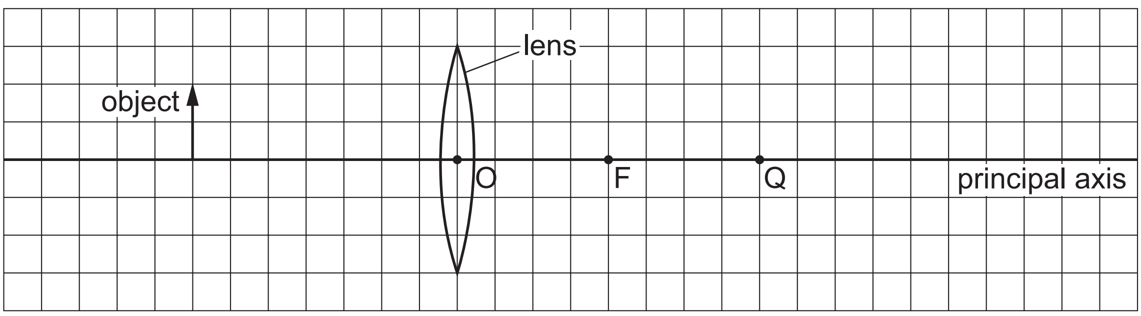
31 The diagram shows light travelling through a medium. The light reaches the boundary with a vacuum as shown. The light emerges travelling along the surface.



What is the refractive index of the medium?

A $\frac{\sin 60°}{\sin 30°}$ **B** $\frac{\sin 60°}{\sin 90°}$ **C** $\frac{\sin 90°}{\sin 30°}$ **D** $\frac{\sin 90°}{\sin 60°}$ **[ON2014/P11/Q21]**

32 The diagram shows an object on the principal axis of a converging (convex) lens. A principal focus of the lens is at F.



Where is the image formed by the lens?

A between O and F

B between F and Q

C at Q

D to the right of Q **[ON2014/P11/Q22]**

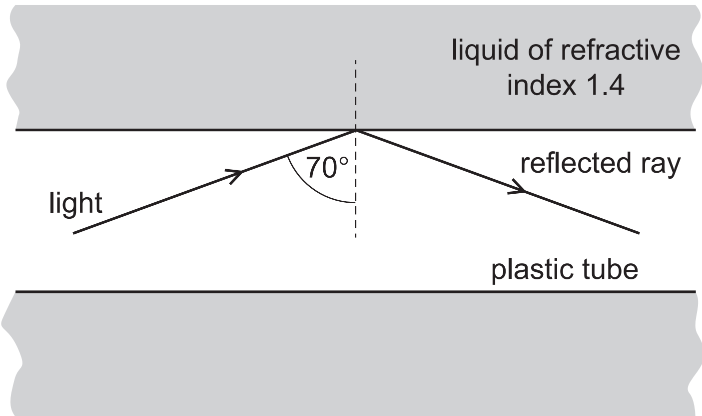
33 A digital camera uses a lens to produce a diminished (reduced in size) image on a light sensor.

Which row shows the correct type of lens and the nature of the image?

	type of lens	nature of image
A	converging	inverted
B	converging	upright
C	diverging	inverted
D	diverging	upright

[ON2014/P11/Q23]

34 A plastic tube is immersed in a liquid of refractive index 1.4. Light travelling in the plastic tube strikes the inside surface at an angle of incidence of 70°. The light undergoes total internal reflection.

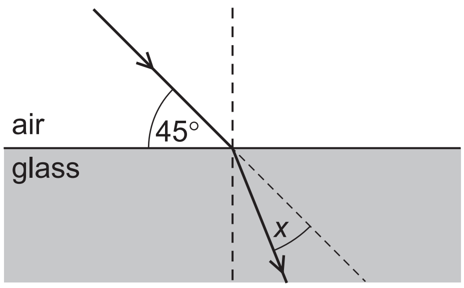


What describes the values of the critical angle in the plastic and the refractive index of the plastic?

	critical angle in plastic	refractive index of plastic
A	greater than 70°	greater than 1.4
B	greater than 70°	less than 1.4
C	less than 70°	greater than 1.4
D	less than 70°	less than 1.4

[ON2014/P12/Q16]

35 A ray of light is incident on the surface of a glass block, as shown in the diagram below.



The refractive index of the glass is 1.5.

The light ray changes direction when entering the glass.

What is the angle x through which the ray moves?

A 30° **B** 28° **C** 17° **D** 15° **[MJ2015/P11/Q22]**

36 An image is formed by a thin converging lens when it is used as a magnifying glass.

What is the correct description of the image?

A real and erect

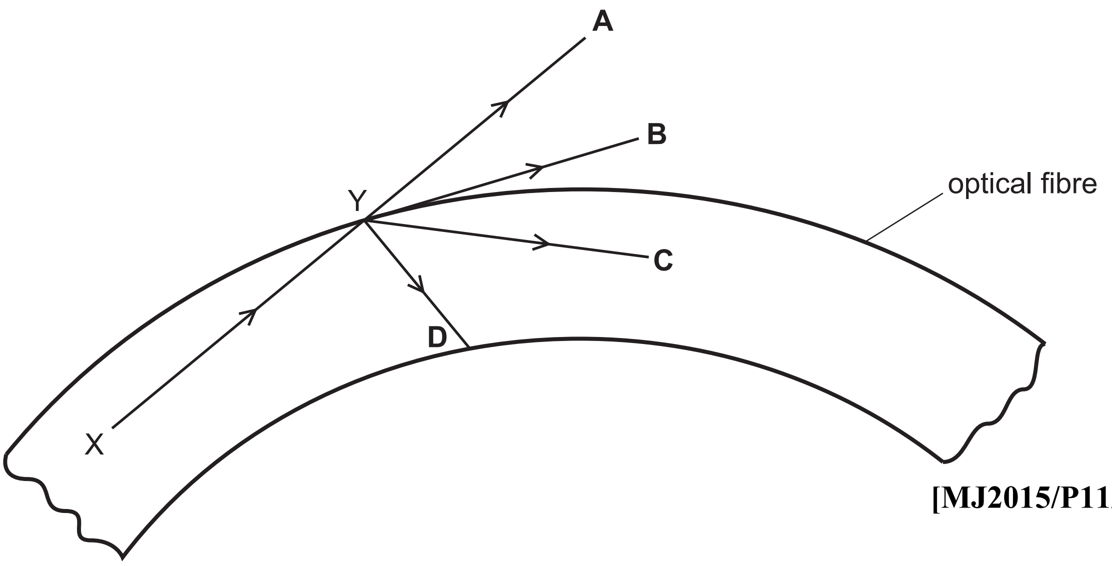
B real and inverted

C virtual and erect

D virtual and inverted **[MJ2015/P11/Q23]**

37 A ray of light travels from X to Y along an optical fibre. The angle of incidence at Y is greater than the critical angle.

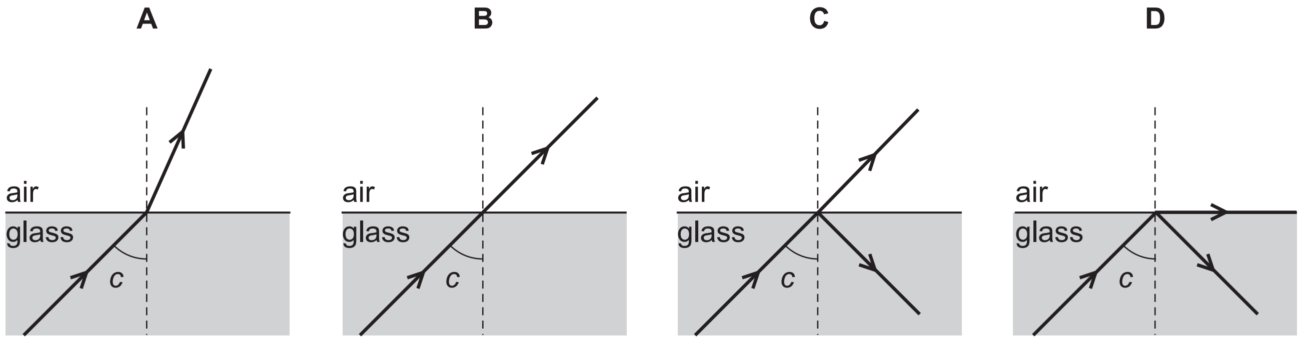
In which direction does the ray of light travel after reaching point Y?



[MJ2015/P11/Q24]

38 A ray of light in glass is incident on the surface at an angle c. The angle c is the critical angle.

Which diagram shows what happens to the light?



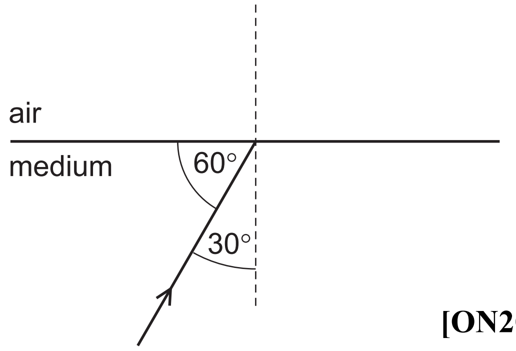
[MJ2015/P12/Q25]

39 A ray of light in a transparent medium of refractive index 1.8 is incident on the surface as shown. The light enters air.

What is the angle between the refracted ray and the normal in air?

A 29° **B** 33°

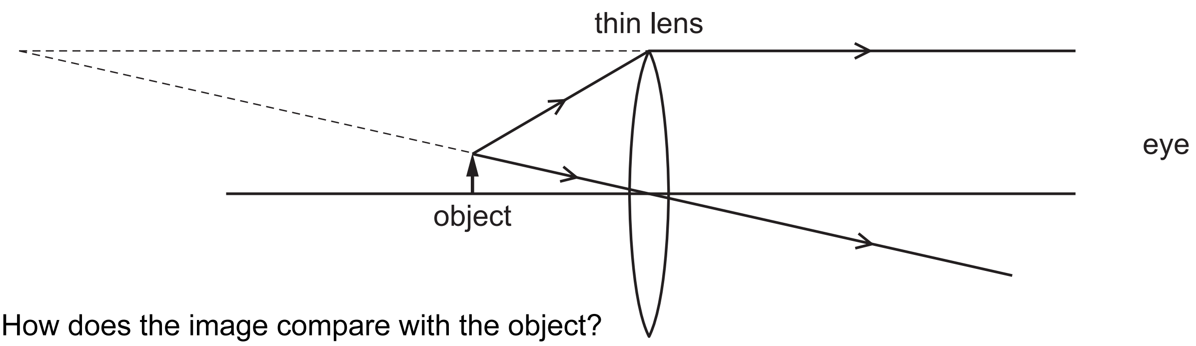
C 54° **D** 64°



[ON2015/P11/Q21]

40 An object is viewed through a converging lens.

The diagram shows the paths of two rays from the top of the object to an eye.



How does the image compare with the object?

A It is larger and inverted.

B It is larger and upright.

C It is smaller and inverted.

D It is smaller and upright.

[ON2015/P12/Q22]

P1 Mark Scheme

ANSWER KEY

1	C	21	B
2	D	22	B
3	D	23	B
4	A	24	B
5	A	25	D
6	A	26	A
7	D	27	B
8	D	28	C
9	D	29	D
10	B	30	A
11	C	31	C
12	A	32	C
13	C	33	A
14	A	34	C
15	D	35	B
16	D	36	C
17	B	37	C
18	B	38	D
19	C	39	D
20	A	40	B

1 **(a)** Describe an experiment to measure the critical angle for light in glass or perspex.

Your answer should include a labelled diagram.

....

....

....

....

....

....

....

....[5]

(b) Fig. 1.1 represents a simple camera.

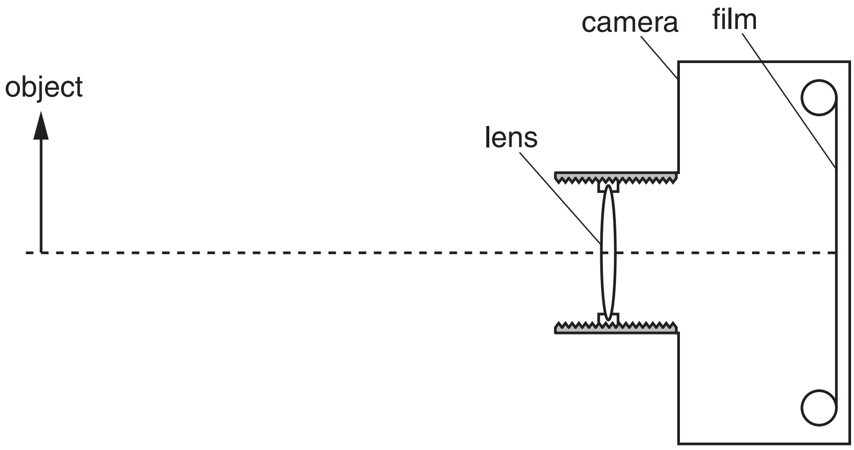


Fig. 1.1 (to scale)

(i) State the type of lens used in this simple camera.

....[1]

(ii) Draw two rays from the top of the object to show how the image is formed on the film. Mark and label the image on the film. [3]

(iii) Define the term *linear magnification*.

....

....[1]

(iv) Fig. 1.1 is drawn to scale. Determine the linear magnification of the object shown in Fig. 1.1.

magnification = ...[1]

(v) Apart from its size, state one other property of the image formed by the lens.

....[1]

(vi) Explain why, when taking photographs of other objects, it may be necessary to move the lens towards the film.

....

....

....

....[3]

[MJ2011/P22/Q10]

2 Fig. 2.1 shows a glass lens in air and its two focal points F_1 and F_2.

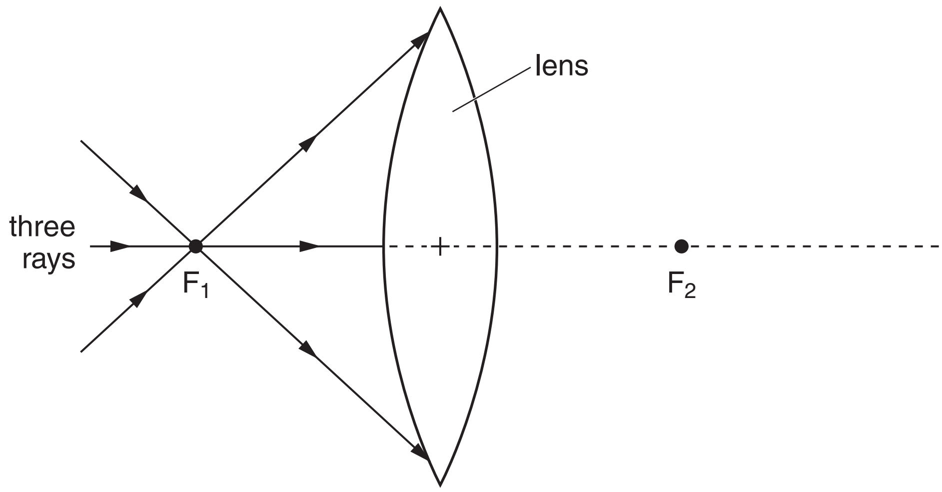


Fig. 2.1

Three rays of light pass through F_1 to the lens.

(a) On Fig. 2.1, continue the three rays through the lens and into the air. [2]

(b) State what happens to the speed of light on

(i) entering the glass lens from air,

....[1]

(ii) leaving the lens and returning to the air.

....[1]

(c) Light of wavelength 6.0×10^{-7} m travels in air at a speed of 3.0×10^{8} m/s.

(i) Calculate the frequency of this light.

frequency = ...[2]

(ii) State the effect, if any, on the frequency as the light enters the glass from air.

....[1]

[ON2011/P21/Q4]

3 Fig. 3.1 shows a short-sighted eye.

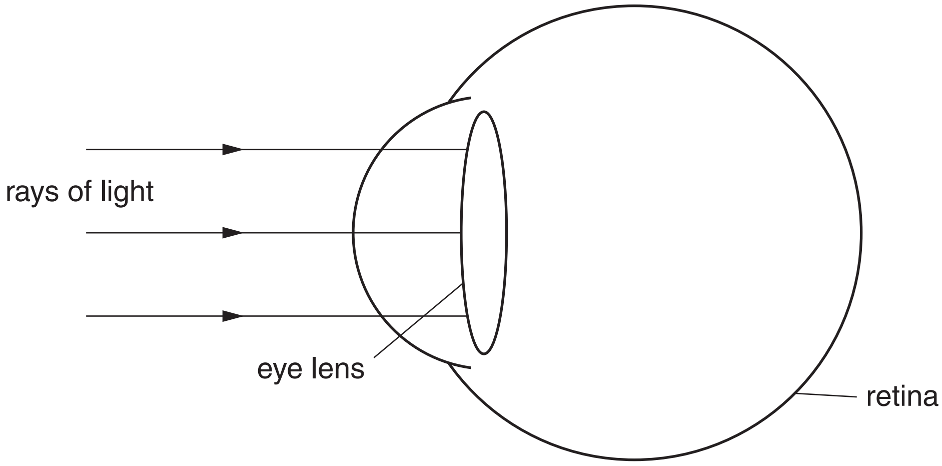


Fig. 3.1

Rays of light from a distant star are parallel as they reach the lens of the eye. Refraction of light as it enters the eye has been ignored in Fig. 3.1.

(a) **(i)** On Fig. 3.1, continue the rays to show their paths inside the short-sighted eye until they strike the retina. [2]

(ii) Explain how your diagram shows that the image of the star seen by the observer is blurred.

....

....[1]

(b) Fig. 3.2 shows three parallel rays of light.

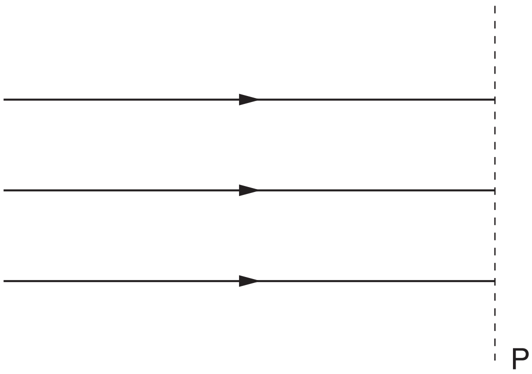


Fig. 3.2

(i) On the line P in Fig. 3.2, draw the shape of a lens that is used to correct short sight. [1]

(ii) On Fig. 3.2, continue the three rays through the lens that you have drawn. [1]

[ON2011/P22/Q8]

4 (a) Fig. 4.1 shows the path of a ray of blue light as it passes through a glass prism.

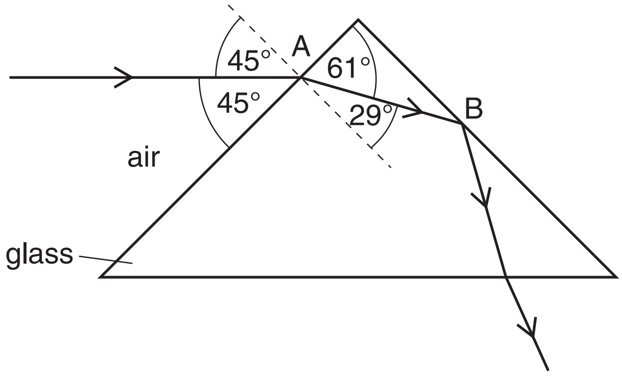


Fig. 4.1

(i) State the wave term used to describe what happens to the ray of light at A.

....[1]

(ii) Using angles from Fig. 4.1, calculate the refractive index of the glass.

refractive index = ...[3]

(iii) Explain why the ray does not emerge from the prism at B.

....

....[2]

(iv) Fig. 4.2 shows a second, horizontal, ray of blue light striking the prism at point C.

On Fig. 4.2, continue the path of the second ray through and out of the glass prism. [2]

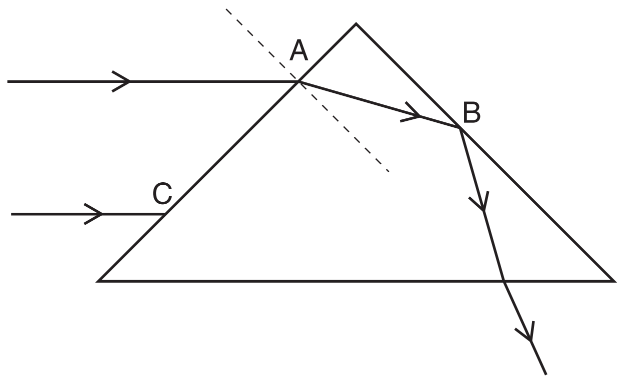


Fig. 4.2

(b) The camera lens shown in Fig. 4.3 is used to photograph the object O.

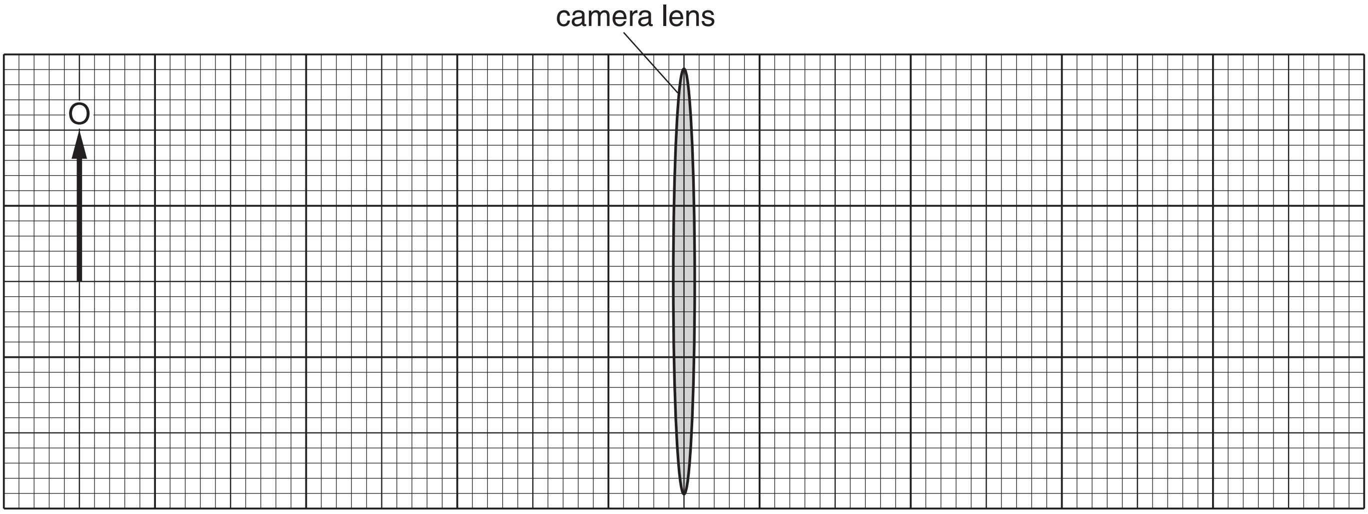


Fig. 4.3 (full scale)

The object O is 2.0 cm high and is placed 8.0 cm from the centre of the lens. The lens has a focal length of 3.0 cm.

(i) Draw rays on Fig.4.3 to find the position and height of the image formed by the lens. Label the image I. [3]

(ii) Determine the height of the image.

....[1]

(iii) The image formed by the lens is a real image.

1. Explain the difference between a real image and a virtual image.

....

....

....[1]

2. Explain how a converging lens is used to produce and view a virtual image.

....

....

....

....[2]

[MJ2012/P21/Q10]

5 Optical fibres are used to transmit telephone signals. Fig. 5.1 shows a ray of light that strikes the inside surface of an optical fibre at P.

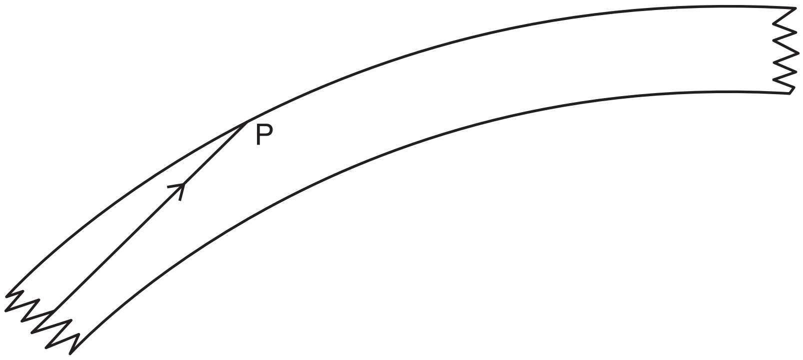


Fig. 5.1

(a) State one advantage of using optical fibres to transmit telephone signals.

....

....[1]

(b) **(i)** On Fig. 5.1, draw a normal at P and mark the angle of incidence with the letter *i*. [1]

(ii) State and explain what happens to the ray at P. Use the term *critical angle* in your answer.

....

....

....[2]

(c) The optical fibre is made of glass of refractive index 1.5.
At the start of the optical fibre, the ray enters the glass from air.
The angle of incidence in the air is 60°.

Calculate the angle of refraction in the glass.

angle = ...[2]

[MJ2012/P22/Q5]

6 A laser produces red light of frequency 4.7×10^{14} Hz. The speed of light in glass is 2.0×10^{8} m/s.

(a) Calculate the wavelength in glass of light from this laser.

wavelength = ...[2]

(b) Describe an experiment to verify *the law of reflection* for light. You may include a diagram in your answer.

....

....

....

....

....

....

....[5]

(c) Fig. 6.1 shows a ray of light travelling in an optical fibre. The ray strikes the side of the fibre at P.

The angle between the ray and the side of the fibre is 7°.

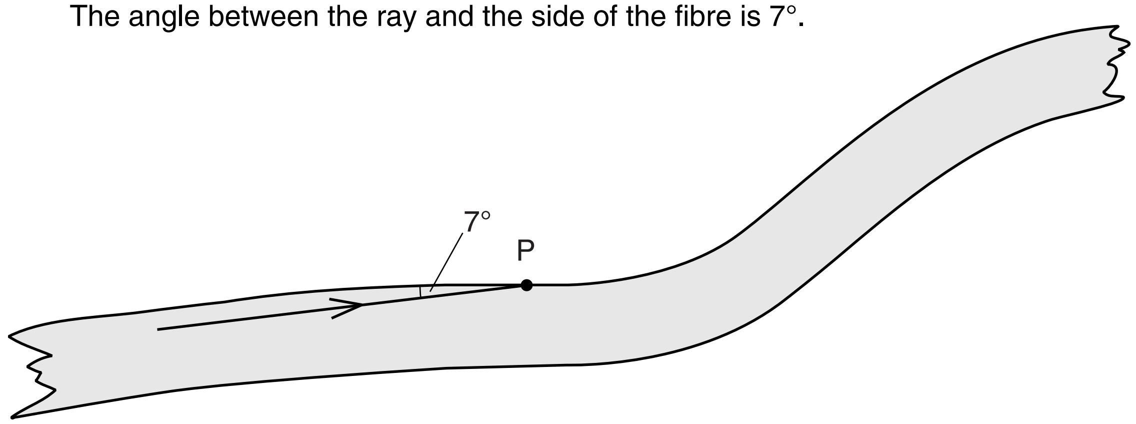


Fig. 6.1

(i) Determine the angle of incidence of the ray at P.

angle = ... [1]

(ii) State and explain what happens to the ray at P.

....

....

.... [2]

(d) A room is illuminated by wall lamps. Fig. 6.2 shows a mirror on the wall behind one of the lamps.

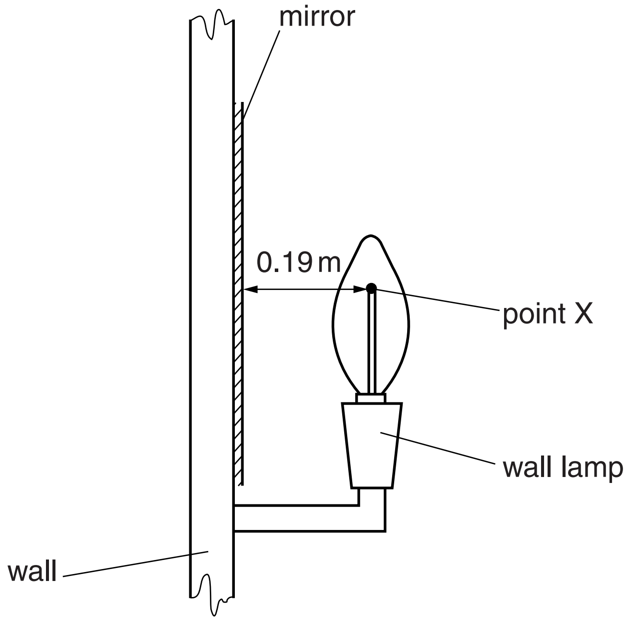


Fig. 6.2 (not to scale)

X is a point on the filament of the lamp. It is 0.19 m in front of the mirror.

(i) On Fig. 6.2, draw rays from X and locate the image of X. Label the image I. [3]

(ii) State the distance between I and the mirror.

distance = ... [1]

(iii) Suggest one advantage of placing a mirror behind the lamp in the room.

....

.... [1]

[ON2012/P22/Q10]

7 Fig. 7.1 shows a ray of light that enters a semicircular glass block at A.
At B, some of the light is reflected and some light leaves the glass and travels along the surface.

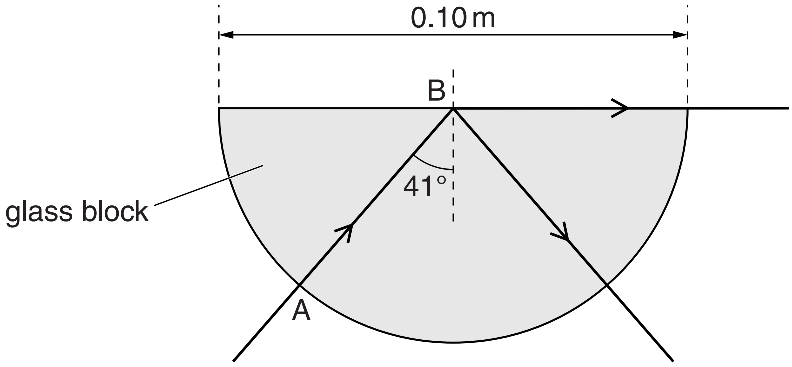


Fig. 7.1

(a) State the name of the angle of incidence marked 41°.

....[1]

(b) Rays of light are incident at B with different angles of incidence.

(i) On Fig. 7.2a, the angle i_1 is less than 41°.
Draw the path taken by the ray of light after B.

(ii) On Fig. 7.2b, the angle i_2 is greater than 41°.
Draw the path taken by the ray of light after B.

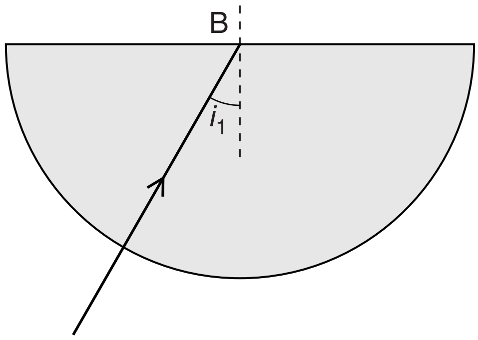


Fig. 7.2a

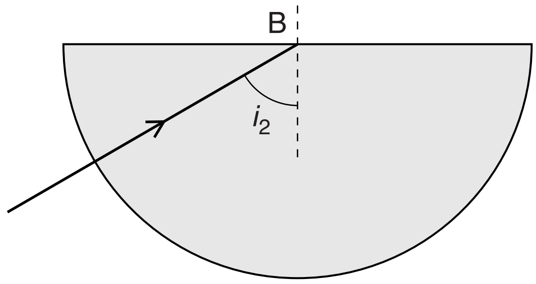


Fig. 7.2b [2]

(c) The speed of light in the glass block is 2.0×10^8 m/s.
The diameter of the glass block is 0.10 m.

Calculate the time taken for the light to travel from A to B.

time = ...[2]

[MJ2013/P21/Q5]

8 A student traces the path of a ray of blue light as it enters and as it leaves a glass prism. Fig. 8.1 shows the trace obtained by the student.

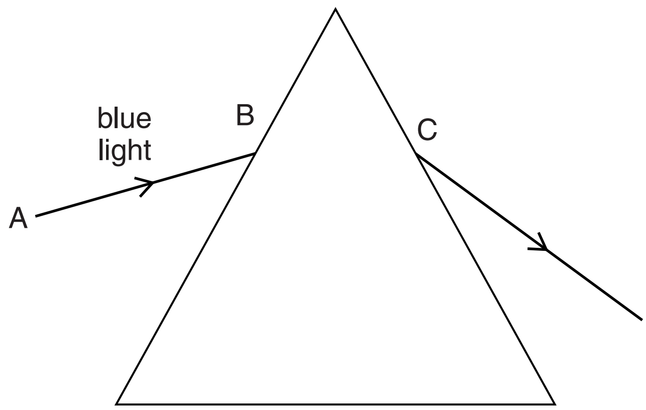


Fig. 8.1

(a) On Fig. 8.1, draw and label, at the point B, the normal, the angle of incidence *i* and the angle of refraction *r*. [3]

(b) State, in terms of the properties of light waves, why the light refracts at B.

....

.... [1]

(c) The angle of incidence for the ray of blue light at B is 45°. The refractive index of the glass is 1.5. Calculate the angle of refraction at B.

angle of refraction = ... [3]

(d) The student performs another experiment with a ray of red light along the line AB.

On Fig. 8.1, show the path taken by this ray of light as it passes through and leaves the prism. [2]

(e) The student performs another experiment with a semicircular glass block and a ray of white light. Fig.8.2 shows the path taken by this ray of light as it enters the glass at P until it hits the straight edge at Q.

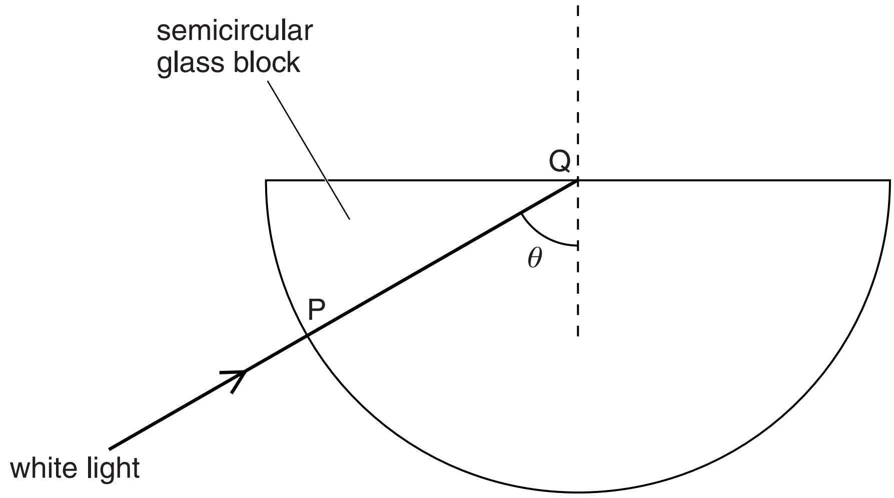


Fig. 8.2

The student finds that there is no change in direction as the ray enters the glass at P and that no light passes out of the glass at Q.

(i) Explain why the ray does not change direction at P.

....

.... [1]

(ii) Explain why no light passes out of the glass at Q.

....

....

.... [2]

(iii) On Fig. 8.2, draw the complete path followed by this ray. [1]

(iv) The student directs the ray of white light into the glass along different paths, so that the angle θ is slowly reduced.

Describe what happens to the ray at Q.

....

....

.... [2]

[MJ2013/P22/Q10]

9 A student goes for a walk in the mountains. During a storm, she sees lightning strike a hillside in the distance. Several seconds later, she hears the thunder caused by the lightning.

(a) **(i)** Explain why she hears the thunder several seconds after she sees the lightning.

....

....[1]

(ii) The student knows the distance to the hillside. She waits for lightning to strike the hillside again. Describe how she can determine a value for the speed of sound in air.

....

....

....

....[2]

(b) After the storm, the student sees a rainbow.

(i) State the speed of light in air.

speed = ...[1]

(ii) Calculate the wavelength in air of light of frequency 7.5×10^{14} Hz.

wavelength = ...[2]

(iii) State the colours of the spectrum in order of increasing wavelength.

....

....[2]

(c) In the laboratory, the student sends blue light into a glass prism placed on a sheet of paper. The arrangement is shown in Fig. 9.1.

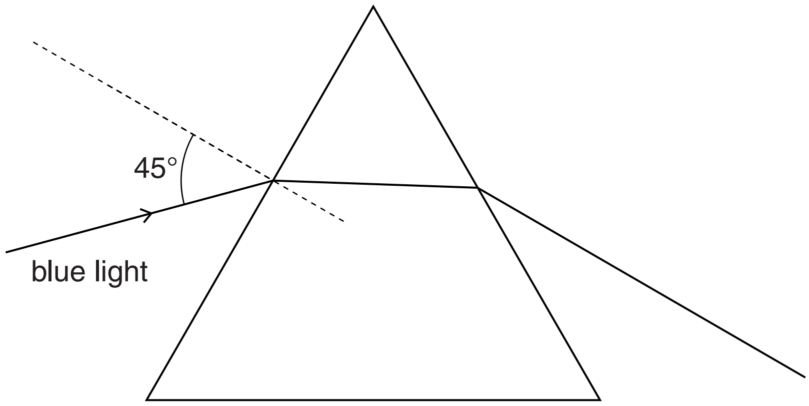


Fig. 9.1

The blue light enters the prism with an angle of incidence of 45°.

(i) On Fig. 9.1, mark the angle of refraction in the glass and label it *r*. [1]

(ii) The student wishes to determine the angle of refraction. On the sheet of paper, she draws the path the ray takes in the glass. Describe how she could do this.

....

....

.... [2]

(iii) **1.** State the formula that relates the angle of incidence *i*, the angle of refraction *r* and the refractive index *n*.

....

.... [1]

2. The angle of refraction in the glass is 28°.

Calculate the refractive index of the glass.

refractive index = ... [1]

(iv) The student sends a ray of red light to the glass prism along the same path as the blue light.

On Fig. 9.1, mark the path taken by the red light after it enters the prism. [2]

[ON2013/P21/Q10]

10 A collector views a postage stamp of height 1.5 cm through a lens. The lens is 2.0 cm from the stamp. The image has a linear magnification of 3.0.

The stamp, the image of the stamp and the position of the lens are shown full scale in Fig. 10.1.

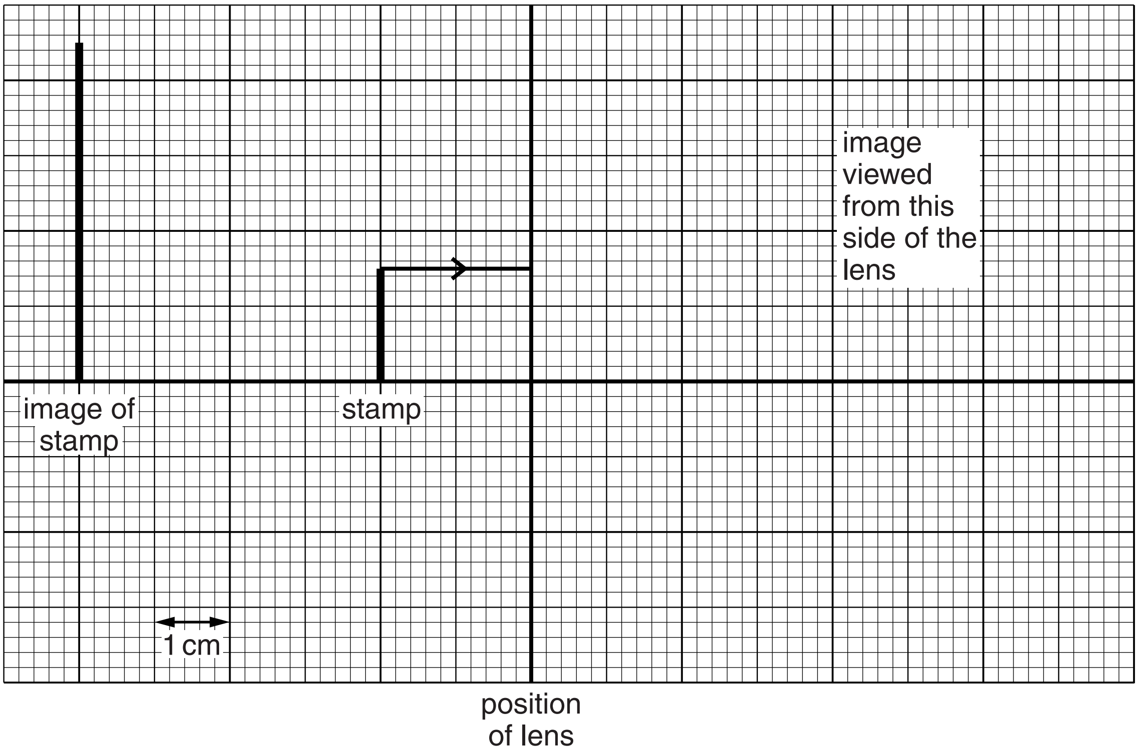


Fig. 10.1 (full scale)

A ray of light from the top of the stamp to the lens is shown on Fig. 10.1.

(a) State the type of lens used.

.... [1]

(b) State what is meant by *linear magnification*.

....

.... [1]

(c) (i) On Fig. 10.1, complete the path of the ray from the top of the stamp after it passes through the lens. [1]

(ii) Use your drawing to determine the focal length of the lens.

.... [1]

(iii) On Fig. 10.1, draw two additional rays from the top of the stamp to show how the image is formed. [1]

[MJ2014/P21/Q3]

11 **(a)** A beam of parallel light strikes a converging lens of focal length 2.8 cm.

The width of the beam before it reaches the lens is 1.0 cm. The width changes on the other side of the lens.

State a distance from the lens where the width of the beam is

(i) less than 1.0 cm,

.... [1]

(ii) more than 1.0 cm.

.... [1]

(b) An object is placed 3.0 cm from a converging lens of focal length of 2.8 cm. Fig. 11.1 is an incomplete, full-scale ray diagram for this arrangement.

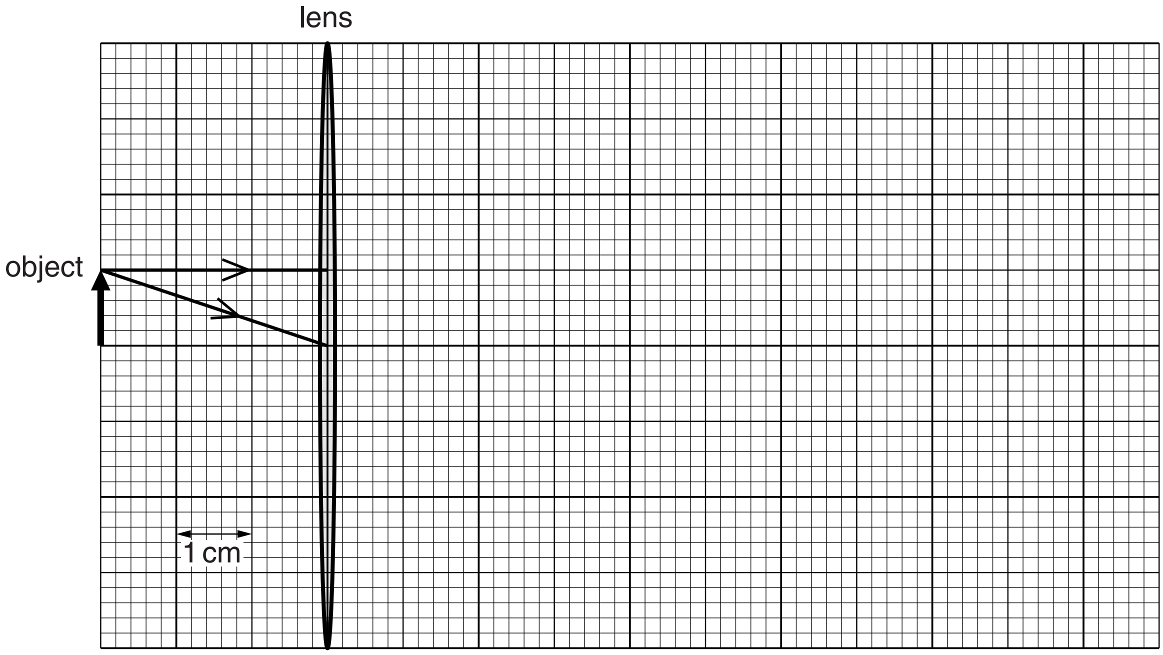


Fig. 11.1 (full scale)

(i) On Fig. 11.1, draw the paths of the two rays after they pass through the lens. [2]

(ii) Explain how your ray diagram shows that the image is more than 11 cm from the lens.

....

.... [1]

(iii) Underline **three** of the following words which describe the image.

diminished inverted magnified real upright virtual [1]

[MJ2014/P22/Q6]

12 **(a)** State the speed of light in air.

....[1]

(b) Fig. 12.1 shows a ray of blue light passing from air into a glass block and refracting at the surface.

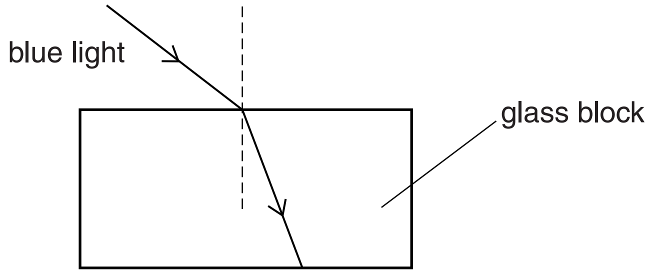


Fig. 12.1 (not to scale)

(i) As the light enters the glass, state what happens to

1. the speed of the light,

....[1]

2. the frequency of the light,

....[1]

3. the wavelength of the light.

....[1]

(ii) On Fig. 12.1, mark and label the angle of incidence *i* and the angle of refraction *r*. [2]

(c) The refractive index of glass for blue light is 1.5.

(i) A ray of blue light strikes the surface of a glass block at an angle of incidence of 89°.

Calculate the angle of refraction of the light in the block.

angle of refraction = ...[3]

(ii) Explain why the angle of refraction of blue light in glass is always less than 45°.

....

....

....

....[2]

(d) Blue light, travelling in air, strikes the side of a different glass block and continues in the same direction as it enters the glass block. Fig. 12.2 shows the ray of light and the shape of the glass block. The critical angle for this glass is 42°.

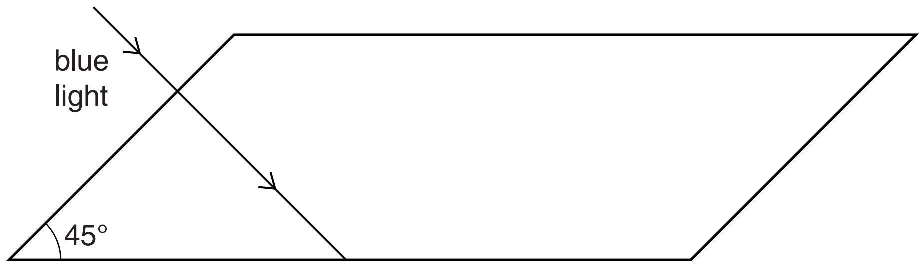


Fig. 12.2

(i) Explain why the light continues in the same direction as it enters the glass block.

....

....

....[2]

(ii) On Fig. 12.2, complete the path of the light until it leaves the glass. [2]

[ON2014/P22/Q10]

13 Fig. 13.1 shows a ray of light entering and passing along an optical fibre.

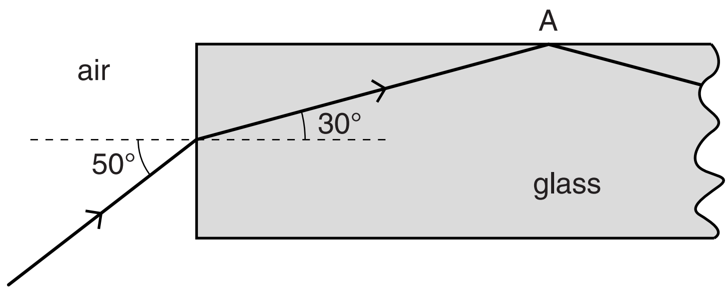


Fig. 13.1

(a) Calculate the refractive index of the glass in the optical fibre.

refractive index = ...[2]

(b) Explain why the ray of light is totally internally reflected at A.

....

....

....[2]

(c) Both optical fibre and copper wire are used to transmit data.

Optical fibre is cheaper and can carry more data per second than copper wire.

State one other advantage of using optical fibre rather than copper wire to transmit data.

....

....[1]

[MJ2015/P21/Q5]

14 A burning candle is placed close to a thin converging lens. The candle acts as the object.

A white, vertical screen is moved to a position on the other side of the lens from the candle.

Fig. 14.1 is a full-scale diagram, on graph paper, of the lens and the screen.

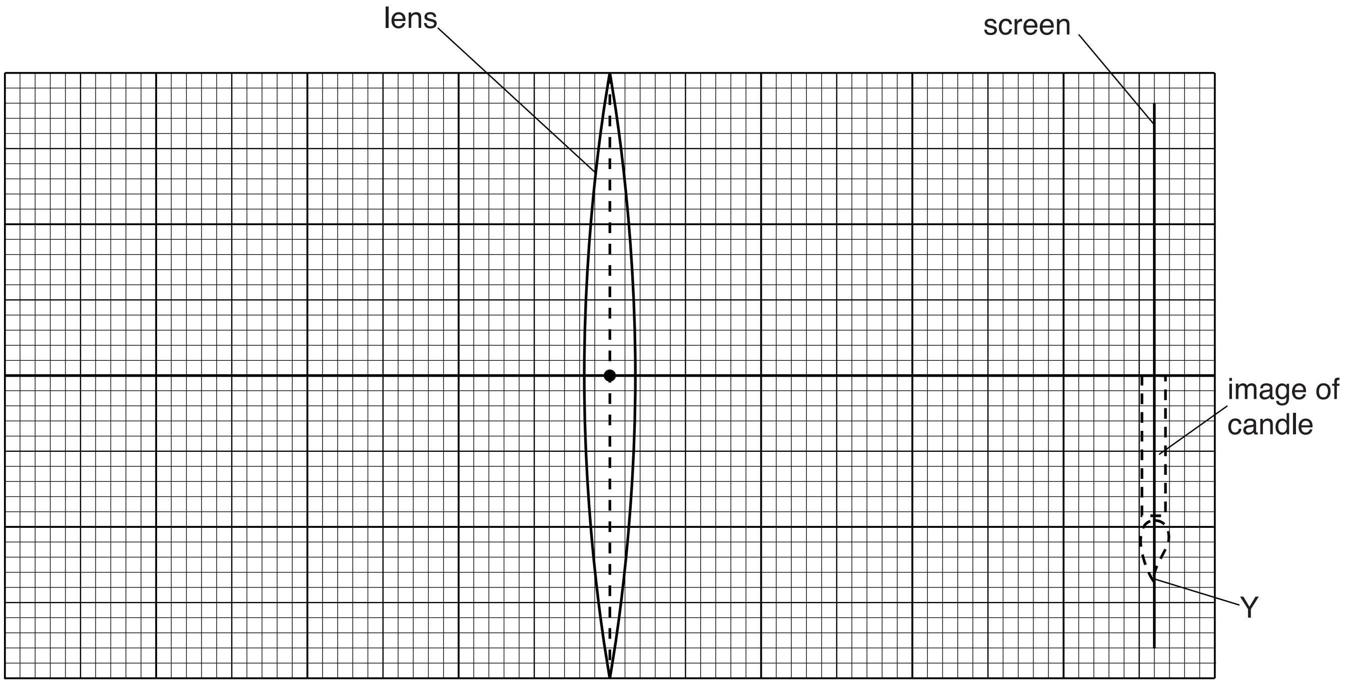


Fig. 14.1

The focal length of the lens is 2.4 cm. The screen is 7.2 cm from the centre of the lens. A sharply focused, inverted image of the candle is produced on the screen, as shown in Fig. 14.1.

(a) Define the term *focal length*.

....

....

.... [1]

(b) **(i)** On Fig. 14.1, mark and label with an F, each of the two focal points (principal foci) of the lens. [1]

(ii) The point Y is the tip of the image.

On Fig. 14.1, draw a ray diagram to locate the position of the top of the object. Label this point X. [3]

(iii) Using Fig. 14.1, determine the distance of the candle from the centre of the lens.

distance = ... [1]

[ON2015/P21/Q6]

15 Light enters a parallel-sided glass block at A. The angle between the side of the block and the ray of light in air is 35°.

The light then strikes the edge of the block at B. Fig. 15.1 shows the ray of light and the glass block.

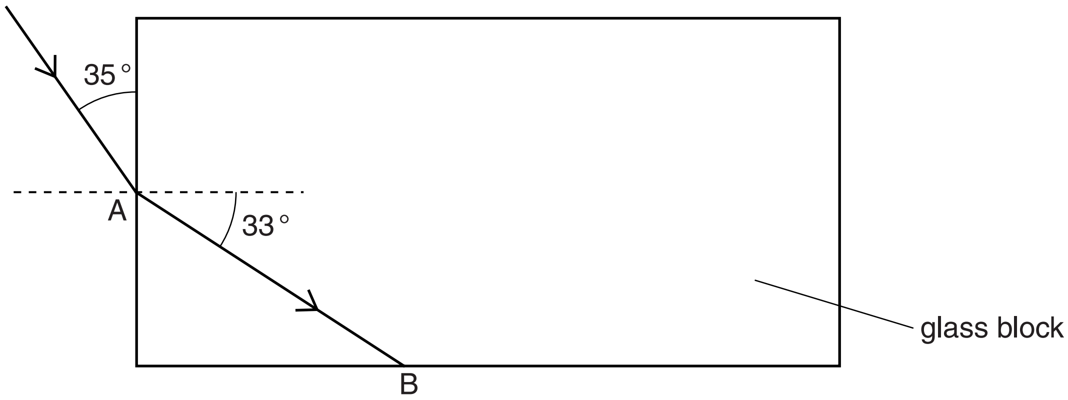


Fig. 15.1

The angle of refraction in the glass at A is 33°.

(a) Calculate the refractive index for light in the glass.

refractive index = ... [2]

(b) The light undergoes total internal reflection at B.

(i) State one condition necessary for total internal reflection to occur.

....

....

.... [1]

(ii) On Fig. 15.1, continue the ray to show the path of the light after total internal reflection at B and until it leaves the block. [1]

[ON2015/P22/Q7]

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10 (a) suitable block (semicircular/rectangular/prism) B1
suitable source of rays (e.g. ray box; pins on incident ray; laser **not** torch)
must be labelled on diagram or clear in text B1
diagram showing incident ray in glass/perspex (no arrow needed)
and correct refraction out into air B1
adjustment of (angle of incidence of) ray until along surface/just no longer emerges B1
(measure) correct angle marked or described clearly or C marked on diagram B1

(b) (i) converging or convex B1

(ii) ray from top through middle of lens undeviated B1
other ray from top of object to same position on film M1
correct image labelled/drawn/marked A1

(iii) ratio of size/height/length/distance of image to size/height/length/distance of object B1

(iv) 0.4(0) (±0.05) no ecf **(iii)** B1

(v) upside down // inverted // real // other side of lens to object // nearer lens than object B1

(vi) (otherwise) not focussed // to/adjust focus // to produce a clear/sharp image // B1
(otherwise) rays do not converge on film // to converge rays onto film //
image on film // object at different distance C1
(otherwise) image formed in front of film // object now further A1 [15]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
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4 (a) one outer ray parallel to principal axis C1
three rays parallel to the principal axis A1

(b) (i) (speed) reduced **or** slows down B1

(ii) (speed) returns to original value/3.0×10^8 m/s B1

(c) (i) ($f =$) c/λ **or** $3.0 \times 10^8 / 6.0 \times 10^{-7}$ C1
$5(.0) \times 10^{14}$ Hz A1

(ii) no effect/unchanged/($f =$) $5(.0) \times 10^{14}$ Hz B1 [7]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
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8 (a) (i) central ray undeviated emerging from lens M1
two outer rays meet the central ray at a point inside the eye **and** carry on to strike the retina A1

(ii) light (from a single point) is spread over an area (on the retina)
or rays do not meet at a point on the retina
or image formed/rays meet/principal focus off retina B1

(b) (i) **any** diverging lens: biconcave, planoconcave, convexoconcave –
i.e. lens **clearly** thinner at the centre B1

(ii) **all** rays diverge B1 [5]

Page 6	Mark Scheme: Teachers' version	Syllabus	Paper
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10 (a) (i) refraction B1

(ii) (n =) sin i/sin r C1
sin 45°/sin 29° C1
1.4585 to more than 1 sig. fig. A1

(iii) the angle of incidence/incident angle is greater than the critical angle B1
total internal reflection occurs B1

(iv) correct refraction at C with ray parallel to AB B1
correct reflection (and correct refraction on other face i.e. downwards) B1

(b) (i) Any TWO of:
undeviated ray through centre of lens
ray parallel to axis through point 3 cm from lens on right after lens
ray through point 3 cm to left of lens parallel to axis after lens M2
rays converge and vertical image drawn and labelled I A1

(ii) 1.2 ± 0.2 cm B1

(iii) 1. real image (can be) formed on screen; virtual image not found on screen; rays converge on real image; rays do not converge on virtual image; rays only appear/seem to come from a point on virtual image B1

2. place object within focal length; between lens and focal point/principal focus B1
view from other side of lens; look through lens; image same side as/behind object B1 [15]

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5 (a) more telephone signals (at one time)
OR great(er) bandwidth; more data (per sec); more signals
OR faster data/information transfer
OR less attenuation; less energy/power/signal loss;
OR long(er) distance (before regeneration)
OR (more) secure
OR less noise/interference OR high(er) quality/clear(er) B1

(b) (i) correct normal and angle marked B1

(ii) total internal reflection B1
angle of incidence is larger than critical angle B1

(c) (n =) sin i/sin r in any form numerical or algebraic C1
35(.2644)° **unit ° needed** A1 [6]

Page 5	Mark Scheme	Syllabus	Paper
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10 (a) (λ =)v/f **or** $2 \times 10^8/4.7 \times 10^{14}$ C1
4.3×10^{-7} m A1 [2]

(b)

raybox/light source/ laser **and** mirror		pin(s) **and** mirror	B1	
shine ray at mirror		place two pins	B1	
mark rays	**or**	two more pins in line with image	B1	
measure i and r **and** equal		measure i and r **and** equal	B1	
repeat		repeat	B1	[5]

(c) (i) 83° B1

(ii) total internal reflection **or** TIR **cao** B1
angle of incidence exceeds critical angle B1 [3]

(d) (i) (at least) one ray from X to mirror M1
(at least) **two** rays from X to mirror **and** correct reflections A1
rays traced back to marked I **or** I marked in correct place (by eye) B1

(ii) 0.19 m B1

(iii) less/no light wasted **or** hall brighter B1 [5]

[Total: 15]

Page 3	Mark Scheme	Syllabus	Paper
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5 **(a)** critical angle B1
(b) **(i)** light refracted out into air **and** bent away from normal (ignore reflected ray) B1
(ii) correct internal reflection (by eye) **and** no refracted ray (**not** at 90°) B1
(c) (t =) distance/speed in any form numerical or algebraic (e.g. d/s, s/v $10/2 \times 10^8$) C1
2.5×10^{-10} s A1 [5]

Page 5	Mark Scheme	Syllabus	Paper
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10 **(a)** correct normal by eye B1
correct angle of incidence between candidate's normal and incident ray B1
correct angle of refraction marked between candidate's normal and BC B1 [3]
(b) decrease / change in speed / wavelength B1 [1]
(c) $n = \sin i / \sin r$ seen in any form C1
($\sin r$ =) $\sin 45° / 1.5$ C1
or 0.47(14) seen
28(.1)° C1 [3]
(d) refracts less at first face and on correct side of normal B1
refraction at second face away from normal so that red ray and blue ray are diverging B1 [2]
(e) **(i)** angle of incidence is 0 B1
or ray along normal/perpendicular to glass
(ii) angle of incidence/θ is larger than critical angle B1
total internal reflection occurs B1
(iii) reflected ray drawn correctly **and** emerging without refraction from block B1
(iv) (eventually) light emerges (into air at Q) B1
or light refracts (out at Q)
or (weak) refracted ray appears
light emerging at Q coloured in some way B1 [6]
or correct description of movement of reflected ray (as θ decreases)

[Total: 15]

Page 4	Mark Scheme	Syllabus	Paper
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10 **(a)** **(i)** speed of sound is (much) less than the speed of light (accept quoted values) B1
(ii) **measure** the time delay (between the lightning and thunder) B1
divide distance by time/delay B1 [3]
(b) **(i)** 3.0×10^8 m/s B1
(ii) (λ =) c/f **or** $3.0 \times 10^8 / 7.5 \times 10^{14}$ C1
4.0×10^{-7} m A1
(iii) (in any order) blue, green, orange, red, yellow, (indigo), (violet) **or** VIBGYOR C1
violet, indigo, blue, green, yellow, orange, red A1 [5]
(c) **(i)** correct angle clear/labelled r B1
(ii) mark/determine entrance and exit points (e.g. trace rays back to glass) B1
join/draw line between entrance and exit points B1
(iii) **1.** $n = \sin i / \sin r$ B1
2. 1.5/1.51/1.506176 with no unit
(**not** just 1.5 without working out) B1
(iv) correct direction of refraction at **both** faces M1
completely correct (above blue) A1 [7]

[Total: 15]

Page 2	Mark Scheme	Syllabus	Paper
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3 **(a)** converging **or** convex [B1]
(b) image height ÷ object height [B1]
(c) **(i)** line when extended back joins top of image with intersection of ray and lens [B1]
(ii) 3.0 ± 0.1 cm ecf from diagram [B1]
(iii) any two further lines from top of stamp that appear to come from the top of the image [B1]

[5]

Page 3	Mark Scheme	Syllabus	Paper
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6 **(a)** **(i)** any single value between 0 and 5.6 cm or a range all of whose values are correct [B1]
(ii) any value beyond 5.6 cm [B1]
(b) **(i)** ray through optical centre undeviated [B1]
other ray correct through or to axis 2.8 cm (± ½ small square) from lens [B1]
(ii) lines drawn meet after 11 cm or rays do not meet (on page) or rays almost parallel [B1]
(iii) inverted, magnified, real all 3 needed and none wrong [B1]

[6]

Page 4	Mark Scheme	Syllabus	Paper
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10 **(a)** 3.0×10^8 m/s B1 **[1]**
(b) **(i)** **1.** decreases **cao** B1
2. no change **cao** B1
3. decreases **cao** B1
(ii) **1.** *i* correctly marked (to normal) B1
2. *r* correctly marked (to normal) B1 **[5]**
(c) **(i)** sin *i* / sin *r* = *n* **or** sin *i* / sin *r* = 1.5 C1
sin 89 / sin *r* = 1.5 **or** sin 89 / 1.5 **or** 0.67(0.666565) C1
42° **or** 41.8025° A1
(ii) *i* equal to / close to 90° **and** *r* less than 45° **or** sin *i* / sin 45 = 1.5 **or** $\sin^{-1}(1/n)/\sin^{-1}(1/1.5)$ **and** 41.8° B1
i never bigger than 89° / 90° **or** sin *i* >1 **or** *r* not be more than *c* B1 **[5]**
(d) **(i)** (sin) *i* = 0 **or** ray enters directly / along normal / perpendicular **or** wavefront / light hits surface all together B1
(sin) *r* = 0 **or** no refraction **or** all slows down together B1
(ii) correct reflection at bottom surface (by eye) M1
second correct reflection at top <u>and</u> no refraction at either point A1 **[4]**

[Total: 15]

Page 2	Mark Scheme	Syllabus	Paper
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5 **(a)** sin *i* / sin *r* or sin 50 / sin 30 C1
1.5(321) A1
(b) moving from more dense to less dense medium **or** moving to lower refractive index (air) B1
angle of incidence is greater than critical angle B1
(c) less heat loss / more efficient B1
less chance of hacking / more secure / less interference
less reduction in signal / less need for boosting / larger distances possible / thinner or less bulky

6	**(a)**	**distance** from (optical) centre to focal point (principal focus)	B1	
	(b) (i)	**both** Fs correctly positioned at ± 1 mm	B1	
	(ii)	**two** of: paraxial ray to lens through focal point to image ray through optical centre ray through focal point and then paraxial to image (**ign.** arrows)	M2	
		X at crossing point of rays	A1	
	(iii)	3.4–3.8 cm	B1	[6]

7	**(a)**	(n =)sin i / sin r **or** sin 55° / sin 33°	M1	
		1.5(040274)	A1	
	(b) (i)	angle of incidence greater than critical angle **or** denser to rarer medium	B1	
	(ii)	reflected ray in correct direction (by eye) to edge of block **and** no second TIR (**ign** marked values)	B1	[4]